CountBCF: Bayesian Causal Forests for Count and Zero-Inflated Count Outcomes

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June 2026

Abstract

Count outcomes, such as hospital admissions, infection incidences, insurance claims, and single-cell read counts, are pervasive in causal research. A significant portion of these outcomes are also zero-inflated, featuring structural zeros from units that never enter the count-generating process. Estimating average and heterogeneous treatment effects on these responses is statistically demanding: traditional parametric count models impose rigid, easily violated covariate-outcome assumptions, while flexible continuous-outcome learners ignore discreteness, mean-variance coupling, and the structural-zero channel. This paper introduces **CountBCF**, a novel, flexible Bayesian model that makes no parametric assumptions on how covariates relate to the outcome. From a single posterior fit, CountBCF delivers both the average treatment effect (ATE) and the per-unit conditional average treatment effect (CATE). In zero-inflated settings, the CATE decomposes in closed form into a count-rate channel and a structural-zero channel, exposing sign-changing heterogeneity that single-channel methods cannot detect. We release the open-source `countbcf` R package, and demonstrate that CountBCF dramatically improves treatment effect estimation, posterior calibration, and interval coverage compared to current benchmark models.

Keywords: Bayesian causal forests; Conditional average treatment effects; Count regression; Heterogeneous treatment effects; Log-linear BART; Poisson and negative binomial regression; Zero-inflated regression; Structural zeros.

1 Introduction

Many of the outcomes that motivate modern causal inference are counts [Choi and Ni, 2023a]. Hospital admissions and emergency-department visits [DEB and TRIVEDI, 1997], infection incidences [Chen et al., 2019], traffic crashes [Shankar et al., 1997], insurance claims [Shi and Valdez, 2012], citation totals [Yan et al., 2011], recidivism events [LAND et al., 1996], and many more are all integer-valued and frequently bunched at zero. The applied literatures that produced these outcomes – epidemiology [Lee and Neocleous, 2010], economics [Winkelmann and Zimmermann, 1994], ecology [Richards, 2007], social sciences [Hilbe, 2025], etc – share an interest not just in whether an intervention works on average: the average treatment effect (ATE), but in for whom it works, and by how much: the conditional average treatment effect (CATE). The past decade of heterogeneous treatment effect (HTE) methodology has answered that demand with flexible nonparametric tools – Bayesian Additive Regression Trees [Chipman et al., 2010, Hill, 2011], Bayesian Causal

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Forests [BCF; Hahn et al., 2020], generalised random forests [Athey et al., 2019], meta-learners [Künzel et al., 2019], and double machine learning [Chernozhukov et al., 2018] – but the bulk of that methodology assumes a continuous, approximately Gaussian response.

Treating counts as if they were Gaussian has three well-documented failure modes [Cameron and Trivedi, 2013, Murray, 2021]. Count distributions place positive mass at zero and at small integers, often on a large fraction of the support; the conditional variance grows with the conditional mean, often super-linearly under overdispersion; and the substantive scientific quantity is almost always a rate or a relative risk, both of which live on the log scale. These pathologies bias point estimates, distort uncertainty quantification, and degrade the heterogeneity signal that HTE methods are designed to recover. A second, sharper failure occurs when the count outcome is also zero-inflated (ZI): a non-trivial fraction of units never enter the count-generating process at all (a structural zero), while the rest accumulate strictly positive counts at heterogeneous intensities. A canonical example is annual healthcare utilisation [Chen et al., 2019]: in many provider panels and insurance cohorts, a sizeable fraction of enrollees record zero visits in a given year because they are effectively out of the care-seeking system, while the remaining enrollees record strictly positive visit counts at heterogeneous intensities. The same pattern arises in manufacturing defect counts, automobile insurance claims, ecological species-abundance surveys, sentencing and recidivism counts, and single-cell RNA sequencing, where a large excess-zero mass reflects technical dropout on top of a count process.

What makes zero-inflation especially awkward for causal inference is that an intervention can move both mechanisms, and the two movements need not point in the same direction [Choi and Ni, 2023b]. Continuing the healthcare example: a workplace wellness programme aimed at preventive care could plausibly pull some enrollees into the system – shrinking the structural-zero fraction – while simultaneously lowering the visit intensity of those already engaged because preventive care substitutes for acute episodes. The response-scale CATE is then a mixture of the two channel effects; whether it ends up positive, negative, or near zero depends on which channel dominates for which unit. A method that ignores the structural-zero mechanism necessarily mis-attributes its movement: if the treatment shifts the structural-zero probability but leaves the count rate untouched, a Poisson-only working model has to absorb the entire change into the rate estimate, biasing it; the same mis-attribution flows in the opposite direction when the treatment moves the count rate but not the structural-zero probability [Choi and Ni, 2023b].

Faced with a count outcome and a CATE question, an applied analyst today reaches for one of three default tools, each of which gives up something the analysis arguably needs. The first is a Gaussian HTE method, most commonly BCF or causal BART, applied to the raw counts as if they were continuous; this preserves the regularisation and uncertainty quantification practitioners trust, but commits all three Gaussian pathologies above. The second is a parametric Poisson, negative binomial, ZIP, or ZINB generalised linear model with treatment-by-covariate interactions [Cameron and Trivedi, 2013, Lambert, 1992, Greene, 1994], possibly stabilised by propensity weighting [Rosenbaum and Rubin, 1983] or a doubly robust correction [Chernozhukov et al., 2018]; this respects the count likelihood but recovers heterogeneity only through pre-specified interaction terms, defeating the nonparametric premise that motivated the move to HTE methods in the first place. The third is a generalised random forest with a Poisson moment condition [Athey et al., 2019], or a related relative-risk-targeted estimator [Kennedy, 2024]; this is honest about the count structure but discards the two structural priors that distinguish BCF from other forest-based HTE methods, leaving heterogeneity estimates exposed to the high-variance behaviour that motivated those priors to begin with. Two-part and hurdle learners [Mullahy, 1986] separate the channels of a zero-inflated outcome but do not share covariate information across them and do not implement BCF differential shrinkage.

The structural priors in question are the distinguishing features of Hahn et al. [2020]. Their construction decomposes the conditional mean of the response into a prognostic function – what the outcome would be in the absence of treatment, as a smooth function of the covariates – and a moderating function – by how much treatment shifts that baseline at each point in covariate space. The two functions are fit by separate BART ensembles with deliberately different priors: the prognostic forest is large and flexible; the moderating forest is small and tightly shrunk toward zero, reflecting the prior belief that treatment effect heterogeneity is typically smoother and weaker than the prognostic surface itself. This differential shrinkage stabilises CATE estimates in finite samples without forcing them to be constant. The second structural ingredient is the inclusion of an estimated propensity score among the inputs to the prognostic forest only. Hahn et al. [2020] show that omitting this control creates regularisation-induced confounding (RIC): the moderating forest absorbs prognostic signal that the regularised prognostic forest underfits, biasing CATE estimates in the direction of the propensity score. Both pathologies are at least as severe under a log-link count likelihood as under a Gaussian one: the multiplicative geometry of the log scale amplifies the consequences of prognostic underfitting, and the count likelihood's tight mean-variance coupling leaves the moderating forest doubly exposed to RIC bias. The same arguments apply per channel in the zero-inflated case.

Our contribution. This paper develops **CountBCF**, a unified Bayesian causal forest framework for count and zero-inflated count outcomes that retains both of BCF's structural ingredients while replacing its Gaussian backbone with a log-linear count likelihood. The framework covers two regimes within a single model and a single sampler:

1. **A count-faithful BCF.** For non-zero-inflated count outcomes, we extend the BCF of Hahn et al. [2020] from a Gaussian to a Poisson and a negative binomial likelihood by routing both the prognostic and the moderating forest through the log-linear BART backbone of Murray [2021]. The result is a BCF that respects discreteness, mass at zero, and mean-variance coupling without abandoning the prognostic/moderating decomposition that defines the BCF family.

2. **A six-forest BCF decomposition for zero-inflated outcomes.** Murray's log-linear BART parameterises a zero-inflated count likelihood through three function groups: the count log-rate f, the structural-zero log-odds f_0, and the not-structural-zero log-odds f_1, with the structural-zero probability identified as $\sigma(f_0 - f_1)$. We replace each group's single forest with the BCF propensity-adjusted prognostic / moderator pair, yielding six forests in total: (μ_f, τ_f) for the count channel and $(\mu_{f_0}, \tau_{f_0}), (\mu_{f_1}, \tau_{f_1})$ for the ZI channel. The non-zero-inflated case is recovered as the three-forest sub-model.

3. **RIC protection on the log scale, per channel.** We retain the propensity-as-covariate fix and the differential-shrinkage prior structure of BCF, with the prognostic forest of each function group receiving more trees and a weaker zero-bias prior than the corresponding moderating forest. The leaf prior is the GIG mixture of Murray [2021], conjugate under a latent-variable augmentation; concentration hyperparameters are calibrated separately for the count and ZI groups, so each moderating forest is shrunk strictly more aggressively than its prognostic counterpart.

4. **Closed-form response-scale heterogeneity and channel decomposition.** The log-linear parameterisation means that, given posterior draws of the forests at any covariate value, the response-scale conditional treatment effect is available without any additional Monte Carlo step: it is a deterministic function of the stored forest values. We report per-unit posterior

means, 50% and 95% credible intervals, and population-averaged ATE summaries, all on the response scale. In the zero-inflated case the same posterior draws produce a per-unit decomposition of the CATE into a count-rate channel $\Delta\lambda(x)$ and a structural-zero channel $\Delta p_{\mathrm{zi}}(x)$.

5. **Open software and a pre-registered simulation study.** We release `countbcf`, an open-source R package built on the `countbart` MCMC backend of Wikle and Zigler [2023], with a single user-facing entry point that consumes a treated/control split and returns the posterior draw matrices needed for the closed-form CATE. The package is accompanied by a pre-registered simulation study (Section 6) that crosses linear and nonlinear data-generating processes with Poisson, NB, ZIP, and ZINB likelihoods and sweeps one-at-a-time over sample size, covariate dimension, and target average treatment effect, scoring all methods against a rate-scale truth.

The remainder of the paper proceeds as follows. Section 2 places CountBCF in the literature on count and zero-inflated causal inference and identifies the specific gap it fills. Section 3 fixes notation, including the latent structural-zero indicator and the response-scale potential-outcome means that are the paper's target estimand. Section 4 reviews the log-linear BART machinery, the GIG-mixture leaf prior, and the redundant (f_0, f_1) parameterisation that makes the ZI leaf updates conditionally conjugate. Section 5 introduces CountBCF proper: the likelihood and parameterisation in both the count and the zero-inflated regimes, the differential-shrinkage prior structure, the augmented Gibbs sampler, and the closed-form response-scale CATE and channel-decomposition summaries. Section 6 registers the simulation study used to evaluate CountBCF against a Gaussian-BCF baseline. Sections 7 and 8 are reserved for the simulation studies results and the application of CountBCF in a real-world setting.

2 Relationship to previous literature

This section reviews the literatures that bear on CATE estimation for count and zero-inflated count outcomes. The cell of the methodological grid in which CountBCF sits – causal $\times$ count-aware (with or without zero-inflation) $\times$ nonparametric Bayesian – is sparsely populated; the subsections that follow walk the adjacent cells in turn before Section 2.4 states the gap that CountBCF fills.

2.1 Causal inference for count outcomes: parametric and semiparametric eras

Causal analysis of count outcomes has its longest tradition in the parametric era of generalised linear models. The Poisson and negative binomial GLMs with a log link [Cameron and Trivedi, 2013] deliver interpretable rate-ratio summaries of the average causal effect under an unconfoundedness assumption and a correctly specified mean function. These models extend cleanly to the negative binomial when equi-dispersion fails, and when unconfoundedness is plausible only conditionally on covariates, the prescription is essentially the same as for continuous outcomes: condition on the covariates or on the propensity score [Rosenbaum and Rubin, 1983]. The two-stage residual inclusion of Terza et al. [2008] and the marginal structural models of Robins et al. [2000] relax the unconfoundedness assumption in different ways while keeping the parametric log-linear-mean structure. Their common feature is that they target the average effect under a fixed functional form: heterogeneity, if permitted at all, enters through pre-specified treatment-by-covariate interactions, which is workable for a small number of binary moderators but not for a moderate-dimensional covariate vector with unknown interaction structure.

The semiparametric and doubly robust literature relaxes the parametric programme's commitment to a single correct outcome model. Funk et al. [2011] survey doubly robust estimators across continuous and discrete outcomes, including Poisson outcome models paired with logistic propensity models. van der Laan and Gruber [2010] and van der Laan and Rose [2011] develop targeted maximum likelihood estimation (TMLE) as a general semiparametric framework whose construction extends naturally to count outcomes via a Poisson or negative binomial submodel for the targeted update step. Chernozhukov et al. [2018] introduce the double / debiased machine learning (DML) framework: the outcome regression and propensity score are fit by arbitrary high-quality machine learners, cross-fitted across folds to avoid overfitting bias, and combined in a Neyman-orthogonal score whose root yields an asymptotically normal ATE estimate. DML is outcome-model-agnostic: a Poisson-loss gradient boosting machine, a neural network, or BART can sit in the outcome slot without changing the orthogonality property. By design these methods target the ATE rather than the per-unit CATE; heterogeneity, when reported at all, is typically delivered by stratifying across pre-specified subgroups or by a separate post-hoc fit of the outcome regression – neither of which preserves the orthogonality property that motivated the doubly robust construction. We adopt the spirit of the doubly robust programme – propensity adjustment of the outcome regression – via the route of Hansen [2008] and Hahn et al. [2020], but reject the restriction to the average effect.

2.2 Classical and causal zero-inflated count models

The starting point for any zero-inflated analysis is the ZIP regression of Lambert [1992], which writes the response as a two-component mixture: with probability $\pi(x)$ the unit is a structural zero, and otherwise it is drawn from a Poisson with rate $\lambda(x)$. The hurdle alternative [Mullahy, 1986] factors the joint density into a binary "any vs. none" model and a zero-truncated count model; hurdle treats all zeros as structural while ZIP allows sampling zeros from the count component as well. Greene [1994] extends both constructions to the negative binomial case. Hall [2000] develops a random-effects ZIP for repeated-measures data, and Li [2012] settles the identifiability question: in the absence of constraints, the (π, λ) pair is not jointly identified from the marginal distribution of Y, but a logistic link on π together with a log link on λ and at least one continuous covariate restore identifiability. The `pscl` package of Zeileis et al. [2008] consolidates the count GLM toolkit – Poisson, NB, ZIP / ZINB, hurdle – in a single `R` treatment.

When interest shifts from describing Y to estimating a treatment effect, the two-component structure of ZI data complicates matters in two ways. First, the response-scale estimand is itself a sum of two interpretable contrasts: a shift in the structural-zero probability and a shift in the count rate. Second, identifying each contrast separately requires that the ignorability assumption be combined with the link-function restrictions of Li [2012] or with an instrument acting on only one channel. The naive plug-in strategy – estimate a ZIP / ZINB regression with treatment-covariate interactions and report the implied mean contrast – inherits all of the distributional rigidity of the underlying GLM. The two-part learner – fit a binary CATE for $\mathbf{1}\{Y_i > 0\}$ and a count CATE for $Y_i \mid Y_i > 0$, then combine – correctly separates the channels but loses joint efficiency across them and does not share covariate information. van der Laan and Rose [2011] discuss how doubly robust / TMLE constructions adapt to these working models; Kennedy [2024] formalises the closely related mismatch between absolute-risk partitioning criteria and relative-risk targets. Guo et al. [2017] develop an IV mediation analysis for ZI counts that defines indirect effect ratios suitable for count outcomes. Yu et al. [2022] propose a Multiplicative Structural Nested Mean Model for ZI nonnegative longitudinal outcomes. Oganisian et al. [2021] build a Bayesian nonparametric model using Dirichlet process mixtures that jointly predict structural zeros, individualised treatment effects, and latent subgroups. Choi and Ni [2023b] approach ZI outcomes from the causal-discovery direction. Lukusa et al. [2017] review the closely related literature on zero-inflated models with missing data. Each of these methods commits either to a particular longitudinal structural form, to a single response-scale estimand, or to a likelihood that does not isolate the structural-zero and count components for inference.

2.3 Heterogeneous-effects methods and the BART line

Methodological work on HTE has been dominated by Gaussian or binary outcomes. Hill [2011] introduced BART [Chipman et al., 2010, 1998] into the causal inference literature as a flexible learner for the outcome regression under unconfoundedness; her construction estimates a single BART model on (X, Z) and reads the CATE off as the difference of the model's predictions at $Z = 1$ and $Z = 0$. This estimator inherits BART's strong empirical performance and produces a fully Bayesian posterior over per-unit treatment effects, but it ties the prognostic and moderating components of the conditional mean to a single tree-structure prior, making no provision for the asymmetric regularisation that distinguishes heterogeneity estimation from outcome estimation, and – applied to a count outcome – it inherits the three Gaussian failure modes catalogued above.

Athey et al. [2019] introduce generalized random forests (GRF), a non-Bayesian alternative built on random forests with locally weighted moment conditions. GRF accommodates a Poisson moment for count outcomes; the asymptotic theory of Wager and Athey [2018] establishes pointwise consistency and asymptotic normality. GRF with a Poisson moment shares a feature CountBCF aims to escape: the splitting criterion targets local likelihood improvement, not relative-risk heterogeneity. Kennedy [2024] formalises this observation. Künzel et al. [2019] catalogue the standard meta-learners (S, T, X, DR), which are estimator-agnostic recipes that delegate the regression and propensity work to a user-chosen base. In principle a meta-learner can be deployed with a count-aware base, but the published meta-learner studies use Gaussian or binary base learners; meta-learners do not by themselves address RIC, and they offer no off-the-shelf solution for ZI counts.

The fourth strand – and the one from which CountBCF draws its computational backbone – is the line of work on BART and its count-aware extensions. Murray [2021] extends BART to count, binomial, and multinomial outcomes by replacing the Gaussian likelihood with a log-linear count likelihood (Poisson or negative binomial) and replacing the Gaussian leaf prior with a generalised inverse Gaussian (GIG) mixture leaf prior whose hyperparameters are calibrated so that, after exponentiation, the implied prior on the forest sum has zero mean and a user-controlled marginal variance. The mixture structure is engineered so that, after a Gamma latent-variable augmentation, the GIG mixture is conjugate. Section 4.2.2 of Murray [2021] extends the construction to zero-inflation through the redundant parameterisation $p_{\mathrm{zi},\,z}(x) = \sigma\big(f_0(x) - f_1(x)\big)$, in which each of f_0 and f_1 carries its own log-linear BART prior, and the data drive symmetry-breaking through latent indicators. The redundancy is precisely what makes the leaf updates conditionally conjugate under a symmetric GIG-mixture prior. Wikle and Zigler [2023] package both constructions into the `countbart` R/C++ implementation that the present paper extends.

Hahn et al. [2020] introduce BCF as the BART-based answer to the CATE-estimation problem under unconfoundedness, with the differential-shrinkage and propensity-as-covariate ingredients catalogued in Section 1. The construction has antecedents in the prognostic score literature of Hansen [2008] and the RIC argument of Hahn et al. [2018]. Wikle and Zigler [2023] restrict their count BART implementation to the predictive setting; they do not pursue the BCF decomposition. The combination of the BCF decomposition with the log-linear count likelihood (with or without zero-inflation), which is the content of the present paper, has not appeared in the literature.

2.4 Identified gap

To summarise: the predictive count and ZI-count cells are well populated, anchored by Murray [2021] and the `countbart` implementation of Wikle and Zigler [2023]. The causal Gaussian cell is well populated, anchored by Hahn et al. [2020] on the Bayesian side and by generalized random forests [Athey et al., 2019, Wager and Athey, 2018] on the frequentist side. The causal count and causal ZI-count cells are sparsely populated, and within them no existing method combines (i) the propensity-adjusted (μ_f, τ_f) decomposition that has made BCF the de-facto Bayesian benchmark for heterogeneous causal effects, with (ii) a count-aware (and where applicable, zero-inflation-aware) likelihood that respects the discreteness, overdispersion, and structural-zero process of integer outcomes. The closest prior work either retains the Gaussian likelihood (BCF; Hahn et al., 2020), drops the BCF decomposition (log-linear BART, with or without ZI; Murray, 2021), separates the two ZI channels and loses joint efficiency (two-part / hurdle learners; Mullahy, 1986; Guo et al., 2017; Yu et al., 2022; Oganisian et al., 2021), or replaces the Bayesian machinery with frequentist random forests targeting an absolute-risk or relative-risk criterion [Athey et al., 2019, Kennedy, 2024]. **CountBCF fills this gap.** It routes the prognostic and moderating forest of each log-linear BART function group through the backbone of Murray [2021] while preserving Hahn et al.'s propensity-adjusted differential-shrinkage prior structure, yielding a count-aware Bayesian causal forest with closed-form per-unit response-scale CATEs, a per-unit channel decomposition in the zero-inflated case, and full posterior uncertainty.

3 Problem statement and notation

Let $Y_i \in \mathbb{Z}_{\geq 0}$ denote a count-valued response, $Z_i \in \{0, 1\}$ a binary treatment indicator, and $X_i \in \mathbb{R}^p$ a length-p vector of pre-treatment covariates. We observe an iid sample $\{(Y_i, Z_i, X_i)\}_{i=1}^n$ from some joint distribution. Capital Roman letters denote random variables, with realisations in lower case. When Y or Z appear without a subscript they denote length-n column vectors; $\mathbf{X}$ is the $n \times p$ design matrix of covariates.

In the potential outcomes framework [Imbens and Rubin, 2015], write $Y_i(0), Y_i(1) \in \mathbb{Z}_{\geq 0}$ for the counterfactual responses under control and treatment, with observed outcome $Y_i = Z_i Y_i(1) + (1 - Z_i) Y_i(0)$ (SUTVA). Throughout we assume strong ignorability:

$$Y_i(0), Y_i(1) \perp\!\!\!\perp Z_i \,\big|\, X_i, \tag{1}$$

$$0 \;<\; \pi(X_i) \;\equiv\; \Pr(Z_i = 1 \mid X_i) \;<\; 1, \tag{2}$$

the second of which (overlap) is required to identify treatment effects everywhere in covariate space. The estimated propensity score is denoted $\widehat{\pi}(X_i)$; for the simulation study of Section 6 we set $\widehat{\pi} = \pi$.

Latent structural-zero indicator. In the zero-inflated regime, the response splits into a structural-zero component and a count component. Let $S_i \in \{0, 1\}$ be the latent indicator that unit i is a structural zero:

$$S_i = 1 \implies Y_i = 0, \qquad S_i = 0 \implies Y_i \mid X_i, Z_i \sim \text{Poisson}(\lambda(X_i, Z_i)) \quad (\text{or NB}(\lambda, \kappa)), \tag{3}$$

and write the conditional structural-zero probability as

$$p_{\text{zi},\,z}(X_i) \;\equiv\; \Pr(S_i = 1 \mid X_i, Z_i = z), \qquad z \in \{0, 1\}. \tag{4}$$

The marginal count-component mean is $\lambda(x, z) = \mathbb{E}[Y_i \mid X_i = x, Z_i = z, S_i = 0]$; we parameterise $\log \lambda$ via a sum-of-trees prior in Section 5. The non-zero-inflated regime is recovered by setting $p_{\text{zi},\,z} \equiv 0$ (no structural-zero process); equivalently, $S_i \equiv 0$ for all i.

Response-scale potential-outcome means. Combining the two channels yields the response-scale potential-outcome means

$$\mu_z(x) \;=\; \mathbb{E}[Y_i(z) \mid X_i = x] \;=\; \big(1 - p_{\mathrm{zi},\,z}(x)\big)\,\exp\!\big(\log\lambda_z(x)\big), \qquad z \in \{0,1\}. \tag{5}$$

Our target estimands are the response-scale CATE and ATE:

$$\tau(x) \;=\; \mu_1(x) - \mu_0(x), \qquad \mathrm{ATE} \;=\; \mathbb{E}_X\big[\tau(X)\big]. \tag{6}$$

When $p_{\mathrm{zi},\,z} \equiv 0$, (5) reduces to $\mu_z(x) = \exp(\log\lambda_z(x))$ and (6) becomes the plain count CATE.

Two-channel decomposition. In the zero-inflated regime, the response-scale CATE in (6) mixes a change in the count intensity and a change in the structural-zero probability. Methods that fit only one channel necessarily miss part of the effect. In Section 5.4 we therefore report – alongside τ – the two per-unit channel decompositions

$$\Delta\lambda(x) \;=\; \exp(\log\lambda_1(x)) - \exp(\log\lambda_0(x)), \qquad \Delta p_{\mathrm{zi}}(x) \;=\; p_{\mathrm{zi},\,1}(x) - p_{\mathrm{zi},\,0}(x). \tag{7}$$

Scales. Treatment effects naturally live on up to three scales:

- the count log-rate scale, on which the moderating forest $\tau_f(x)$ is parameterised; this is a relative-risk-like quantity, $\tau_f(x) \approx \log\{\lambda_1(x)/\lambda_0(x)\}$ in the non-zero-inflated case;
- the ZI log-odds scale, on which the treatment shift in η_{zi} is parameterised (zero-inflated regime only);
- the response (rate) scale, $\tau(x)$ in (6), against which all reported metrics are evaluated.

All relevant quantities are reported. The simulation study (Section 6) is scored against the rate-scale truth.

4 Background: log-linear BART

This section gives the condensed restatement of the log-linear BART construction of Murray [2021] – including the zero-inflation extension of §4.2.2 – that we need to define CountBCF in Section 5. The prognostic and moderating forests of each function group are individually log-linear BART regressions, so the leaf prior, the conditional sufficient statistics, and the latent-variable augmentation recapped here reappear unchanged inside the CountBCF Gibbs sampler. A reader already fluent in Murray [2021] §§3–4.2.2 may skip to Section 5.

4.1 The BART prior

A single BART regression [Chipman et al., 2010] expresses an unknown function $f : \mathbb{R}^p \to \mathbb{R}$ as a sum of m shallow regression trees,

$$f(\mathrm{x}) \;=\; \sum_{h=1}^{m} g(\mathrm{x};\; T_h, M_h), \tag{8}$$

where T_h is a binary tree partitioning $\mathbb{R}^p$ and $M_h = (\mu_{h,1}, \ldots, \mu_{h,b_h})'$ collects the leaf parameters at the b_h terminal nodes of T_h; the operator $g(\mathrm{x};\; T_h, M_h)$ returns $\mu_{h,t}$ when x falls in terminal node t of T_h. The tree prior of Chipman et al. [1998] assigns each internal node at depth d a split probability $\alpha\,(1+d)^{-\beta}$, with $(\alpha, \beta) = (0.95, 2)$ biasing trees toward stubs. With a Gaussian response, a Gaussian leaf prior is conjugate, and the posterior is sampled by a backfitting Gibbs scheme that updates one tree at a time using grow / prune / change / swap MH moves on T_h and a conjugate draw of $M_h \mid T_h$.

4.2 The log-linear extension

For non-negative responses one writes the same sum-of-trees representation on the log scale,

$$\log f(\mathrm{x}) \;=\; \sum_{h=1}^{m} g(\mathrm{x};\; T_h, M_h), \tag{9}$$

and reparameterises each leaf parameter as $\lambda_{h,t} = \exp(\mu_{h,t}) > 0$, so that $f(\mathrm{x}) = \prod_h \lambda_{h,t(\mathrm{x};T_h)}$. The Poisson and negative binomial regression models take the form

$$Y_i \mid \mathbf{X}_i \;\sim\; \mathrm{Poisson}\big(\mu_{0i}\, f(\mathbf{X}_i)\big), \qquad Y_i \mid \mathbf{X}_i \;\sim\; \mathrm{NB}\big(\mathrm{size} = \kappa,\; \mathrm{mean} = \mu_{0i}\, f(\mathbf{X}_i)\big), \tag{10}$$

with μ_{0i} a unit-specific offset and κ the NB dispersion, satisfying $\mathrm{Var}(Y_i) = \mu_{0i} f(\mathbf{X}_i) + \mu_{0i}^2 f(\mathbf{X}_i)^2/\kappa$. The Poisson model is the $\kappa \to \infty$ limit. Both likelihoods reappear inside the zero-inflated construction of Section 4.5: the not-structural-zero component of the ZI mixture is exactly the model in (10), while the structural-zero process is governed by an independent pair of log-linear BART components introduced below.

After collecting all but the hth tree's fit into $f^{-h}(\mathbf{X}_i) = \prod_{l \neq h} g(\mathbf{X}_i; T_l, \Lambda_l)$, the contribution of a single observation to the joint posterior over (T_h, Λ_h) has the conditionally Gamma-shaped form

$$L\big((T_h, \Lambda_h);\; T_{-h}, \Lambda_{-h}, y\big) \;\propto\; \prod_{t=1}^{b_h} \lambda_{h,t}^{r_{h,t}} \,\exp\!\big[-s_{h,t}\, \lambda_{h,t}\big], \tag{11}$$

with conditional sufficient statistics

$$r_{h,t} \;=\; \sum_{i:\, \mathbf{X}_i \in \mathcal{A}_{h,t}} u_i, \qquad s_{h,t} \;=\; \sum_{i:\, \mathbf{X}_i \in \mathcal{A}_{h,t}} v_i\, f^{-h}(\mathbf{X}_i), \tag{12}$$

$\mathcal{A}_{h,t}$ the leaf-t partition cell of T_h, and (u_i, v_i) likelihood-specific summaries that depend on the function group (see Sections 4.4 and 4.5).

4.3 The GIG-mixture leaf prior

Murray [2021] constructs a leaf prior conjugate to (11) from two pieces. The generalised inverse Gaussian (GIG) family has density

$$p_{\mathrm{GIG}}(\lambda \mid \eta, \chi, \psi) \;\propto\; \lambda^{\eta-1} \exp\!\Big[-\tfrac{1}{2}\big(\chi/\lambda + \psi\lambda\big)\Big], \qquad \lambda > 0, \tag{13}$$

covering the Gamma ($\chi = 0$) and inverse-Gamma ($\psi = 0$) families as boundary cases. The CountBCF leaf prior is the equal mixture

$$p\big(\lambda_{h,t} \mid c, d\big) \;=\; \tfrac{1}{2}\, p_{\mathrm{GIG}}\big(\lambda_{h,t} \mid -c,\, 2d,\, 0\big) \;+\; \tfrac{1}{2}\, p_{\mathrm{GIG}}\big(\lambda_{h,t} \mid +c,\, 0,\, 2d\big), \tag{14}$$

with stochastic representation $\lambda_{h,t} = W$ with probability 1/2 and $\lambda_{h,t} = 1/W$ otherwise, $W \sim \mathrm{Gamma}(c, d)$. This construction makes $\log \lambda_{h,t}$ exactly symmetric about zero and keeps (14) conjugate to (11): the leaf full conditional is again a GIG mixture, and the integrated likelihood for the tree MH step is a ratio of GIG normalising constants [Murray, 2021, §3.1].

The pair (c, d) is reparameterised by a user-facing concentration – a_0 for the count component, z_{conc} for the ZI components (Section 4.5) – and the number of trees m, by setting $\psi_1(c) = a_0^2/m$ and $d = \exp(\psi(c))$, where ψ and ψ_1 are the digamma and trigamma functions. This calibration gives $\mathbb{E}[\log \lambda_{h,t}] = 0$ and $\mathrm{Var}(\log \lambda_{h,t}) = a_0^2/m$, so the implied prior on $\log f(\mathbf{X}) = \sum_h \log \lambda_{h,t(\mathbf{X};T_h)}$ is approximately $\mathrm{N}(0, a_0^2)$ at any $\mathbf{X}$. The concentration therefore plays the same role for $\log f$ that the prior leaf scale plays for f in Gaussian BART. For large m or small a_0 the calibration admits the closed-form approximation $c \approx m/a_0^2 + 1/2$, $d \approx m/a_0^2$ [Murray, 2021, Proposition 3].

4.4 Latent-variable augmentation for the count component

To put the count likelihood into the form (11), Murray [2021] introduces one Gamma latent variable ξ_i per observation, with full conditional $\xi_i \mid - \sim \text{Gamma}(\kappa + y_i,\ \kappa + \mu_{0i} f(\mathbf{X}_i))$ under the NB likelihood, and $\xi_i \equiv 1$ under the Poisson likelihood. Conditioning on ξ_i, the per-observation contribution of f collapses to a Poisson-type kernel $f(\mathbf{X}_i)^{y_i} \exp[-\xi_i \mu_{0i} f(\mathbf{X}_i)]$, which integrated against the GIG-mixture prior (14) gives (11) with $u_i = y_i$ and $v_i = \xi_i \mu_{0i}$. Under the NB likelihood the dispersion κ is updated by a random-walk MH step on $\log \kappa$ with a Beta-prime prior implied by $\kappa/(1+\kappa) \sim \text{Beta}(\kappa_a, \kappa_b)$. We defer the specific update to Section 5.3.

4.5 Zero-inflation and the redundant (f_0, f_1) parameterisation

For a zero-inflated count outcome the likelihood is a two-component mixture: with conditional probability $\omega(\mathbf{X}_i)$ the response is a structural zero, and otherwise it is drawn from the Poisson or NB law (10). Murray [2021] §4.2.2 writes the mixing probability through a pair of log-linear BART forests f_0, f_1,

$$\omega(\mathbf{X}_i) \;=\; \frac{\exp(f_0(\mathbf{X}_i))}{\exp(f_0(\mathbf{X}_i)) + \exp(f_1(\mathbf{X}_i))} \;=\; \sigma\big(f_0(\mathbf{X}_i) - f_1(\mathbf{X}_i)\big), \tag{15}$$

where each of f_0, f_1 has its own log-linear BART prior (9) with an independent GIG-mixture leaf prior (14) (with concentration z_{conc}). The parameterisation is redundant: adding a common constant to both f_0 and f_1 leaves ω unchanged. The redundancy is identified through the prior – both forests are centred at zero with symmetric GIG-mixture leaf priors – and confers two concrete benefits. Conditionally Gamma-shaped leaf updates with closed-form GIG-mixture posteriors are obtained via a pair of latent variables. Let $z_i \in \{0,1\}$ be the augmented mixture indicator that resolves whether an observed $Y_i = 0$ came from the structural process ($z_i = 0$) or from the count process with a zero draw ($z_i = 1$). When $Y_i > 0$, $z_i \equiv 1$; when $Y_i = 0$, z_i has the Bernoulli full conditional

$$\Pr(z_i = 1 \mid -) \;=\; \frac{(1-\omega(\mathbf{X}_i))\, p_{\text{count}}(0 \mid \mathbf{X}_i)}{\omega(\mathbf{X}_i) + (1-\omega(\mathbf{X}_i))\, p_{\text{count}}(0 \mid \mathbf{X}_i)}, \tag{16}$$

where $p_{\text{count}}(0 \mid \mathbf{X}_i)$ is the count-component zero probability ($\exp(-\lambda)$ for Poisson / ZIP, $(\kappa/(\kappa+\lambda))^{\kappa}$ for NB / ZINB). Let further $\phi_i > 0$ be an Exponential latent decoupling f_0 and f_1, with full conditional

$$\phi_i \;\mid\; - \;\sim\; \text{Exponential}\big(\exp(f_0(\mathbf{X}_i)) + \exp(f_1(\mathbf{X}_i))\big). \tag{17}$$

Conditioning on (z_i, ϕ_i), the joint conditional likelihood of (f_0, f_1) factorises into independent terms of the form (11): the f_0 leaves have $(u_i, v_i) = (1 - z_i, \phi_i)$ and the f_1 leaves have $(u_i, v_i) = (z_i, \phi_i)$ [Murray, 2021, §A.5, Proposition 1]. The redundant sampler is also numerically better behaved than a strictly identified parameterisation: Murray [2021, Supplement §A] report substantially better effective-sample-size performance across the full probability range. In the body of this paper we write the structural-zero log-odds as $\eta_{\text{zi}} = f_0 - f_1$ and report inference for the identified functionals only.

4.6 Implementation backbone

We inherit the C++ implementation of the log-linear BART sampler – the GIG-mixture leaf draw, the grow / prune / change / swap tree MH moves, the latent-variable updates (z_i, ξ_i, ϕ_i), the soft-mixture-weight recomputation, and the dispersion MH step – from the `countbart` R package of Wikle and Zigler [2023]. CountBCF adds the BCF (μ_f, τ_f) decomposition on top of this backend by replacing the single forest of log-linear BART (per function group) with a forest pair and introducing a per-observation weight that routes treated and control units to the correct forest; we describe that change next.

5 CountBCF: BCF decomposition for count and ZI count outcomes

This section defines CountBCF. Section 5.1 states the likelihood and parameterisation in both the count and the zero-inflated regimes; Section 5.2 collects the priors and the differential shrinkage applied symmetrically across function groups; Section 5.3 writes out the MCMC sweep in Algorithm 1 and the associated Gibbs and MH updates; Section 5.4 explains how posterior draws are post-processed into the estimands of Section 3, including the per-unit channel decomposition in the ZI case.

5.1 Likelihood and parameterisation

CountBCF models the conditional mean of a count response (and, in the ZI regime, the structural-zero probability) on the log scale using a treatment-aware BCF decomposition. We treat the count and zero-inflated cases as two sub-models of a common framework, selected by the user via a `count_model` switch that takes values in {`poisson`, `nb`, `zipoisson`, `zinb`}.

Count regime: three forests. For unit i with covariates X_i and treatment indicator $Z_i \in \{0, 1\}$, the Poisson / NB sub-model writes

$$\boxed{\log \lambda(X_i, Z_i) \;=\; \mu_f\big(X_i,\, \widehat{\pi}(X_i)\big) \;+\; Z_i \cdot \tau_f(X_i),} \tag{18}$$

with response

$$Y_i \mid X_i, Z_i \;\sim\; \text{Poisson}\big(\lambda(X_i, Z_i)\big) \;\text{ or }\; \text{NB}\big(\text{size} = \kappa,\; \text{mean} = \lambda(X_i, Z_i)\big), \tag{19}$$

where the size parameter $\kappa > 0$ governs overdispersion. The Poisson model is the $\kappa \to \infty$ limit. Decomposition (18) is the count analogue of the additive BCF decomposition $\mathbb{E}[Y \mid X, Z] = \mu_f(X, \widehat{\pi}) + Z\tau_f(X)$ of Hahn et al. [2020]. Both pieces are log-linear BART forests in the sense of Section 4.2: μ_f is a sum of m_μ trees with leaf parameters drawn from the GIG mixture (14), and similarly τ_f is a sum of m_τ trees. The prognostic forest models the control-arm log-mean across the whole sample; the moderating forest enters only for treated units and captures treatment-effect heterogeneity. As in BCF, $\widehat{\pi}$ is appended to the covariates feeding μ_f but not those feeding τ_f.

Zero-inflated regime: six forests, three groups. The ZIP / ZINB sub-model attaches a BCF (μ, τ) decomposition to each of the three log-linear BART forests f, f_0, f_1 of Section 4.5. We index the three function groups by $g \in \{0, 1, 2\}$ with $g = 0$ the count log-rate, $g = 1$ the structural-zero log-odds component f_0, and $g = 2$ the not-structural-zero log-odds component f_1. Then

$$\boxed{\begin{aligned} \log \lambda(X_i, Z_i) \;&=\; \mu_f\big(X_i,\, \widehat{\pi}(X_i)\big) \;+\; Z_i \cdot \tau_f(X_i), \\ \eta_{\text{zi}}(X_i, Z_i) \;&=\; \big[\mu_{f_0}\big(X_i,\, \widehat{\pi}(X_i)\big) + Z_i \cdot \tau_{f_0}(X_i)\big] \\ &\qquad - \big[\mu_{f_1}\big(X_i,\, \widehat{\pi}(X_i)\big) + Z_i \cdot \tau_{f_1}(X_i)\big], \\ p_{\text{zi},\, z}(X_i) \;&=\; \sigma\big(\eta_{\text{zi}}(X_i,\, Z_i = z)\big), \qquad z \in \{0, 1\}, \end{aligned}} \tag{20}$$

Table 1: The six forests of CountBCF in the zero-inflated regime. The function group g labels the log-linear BART target inherited from Murray [2021]; the BCF role s labels the prognostic (μ) vs. moderating (τ) piece introduced by the (μ, τ) decomposition. The non-ZI sub-model retains only the $g = 0$ row.

g	Function group	Forest pair $(\mu^{(g)}, \tau^{(g)})$	Trees (m_μ, m_τ)	Weight ω_i on $\tau^{(g)}$
0	Count log-rate $\log \lambda$	$(\mu_f,\ \tau_f)$	(250, 50)	Z_i
1	ZI log-odds f_0	$(\mu_{f_0},\ \tau_{f_0})$	(100, 50)	Z_i
2	not-ZI log-odds f_1	$(\mu_{f_1},\ \tau_{f_1})$	(100, 50)	Z_i

together with the ZI count likelihood $Y_i \mid S_i = 1 = 0$, and $Y_i \mid S_i = 0, X_i, Z_i \sim \text{Poisson}(\lambda(X_i, Z_i))$ or $\text{NB}(\kappa, \lambda(X_i, Z_i))$, where $\Pr(S_i = 1 \mid X_i, Z_i) = p_{\text{zi},\, z}(X_i)$. The non-ZI sub-model collapses (20) by dropping the η_{zi} line and the two ZI groups, recovering (18). Equation (20) contains six unknown functions, two per function group, summarised in Table 1. Each is itself a log-linear BART regression: a sum of $m_{s,g}$ shallow trees with leaf parameters drawn from the GIG mixture, with $s \in \{\mu, \tau\}$ indexing the BCF role and $g \in \{0, 1, 2\}$ indexing the function group. For each group, the prognostic forest $\mu^{(g)}$ enters with weight $\omega_i = 1$; the moderating forest $\tau^{(g)}$ enters with weight $\omega_i = Z_i$. The propensity score estimate $\widehat{\pi}$ is appended to the covariates of all three prognostic forests but not to the moderating ones.

Scales recovered. On the count log-rate scale, $\tau_f(x)$ is itself a CATE: the conditional log relative rate (of the not-structural-zero sub-population in the ZI case), $\tau_f(x) = \log\{\lambda_1(x)/\lambda_0(x)\}$. On the ZI log-odds scale, the treatment shift in the structural-zero log-odds is

$$\Delta\eta_{\text{zi}}(x) \;\equiv\; \eta_{\text{zi}}(x, 1) - \eta_{\text{zi}}(x, 0) \;=\; \tau_{f_0}(x) - \tau_{f_1}(x), \tag{21}$$

so τ_{f_0}, τ_{f_1} are jointly identified up to a common shift – only the difference is statistically relevant for the ZI channel. On the response (rate) scale, the potential-outcome means (5) are

$$\mu_z(x) \;=\; \big(1 - \sigma(\eta_{\text{zi}}(x, z))\big)\, \exp\!\big(\mu_f(x, \widehat{\pi}(x)) + z\, \tau_f(x)\big), \qquad z \in \{0, 1\}, \tag{22}$$

and $\tau(x) = \mu_1(x) - \mu_0(x)$. In the non-ZI sub-model $p_{\text{zi},\, z} \equiv 0$ and (22) reduces to $\mu_z(x) = \exp(\mu_f + z\tau_f)$, giving the closed-form count CATE

$$\tau(x) \;=\; \exp\!\big(\mu_f(x, \widehat{\pi}(x))\big)\, \big(\exp(\tau_f(x)) - 1\big), \tag{23}$$

which makes explicit that the absolute treatment effect on the rate scale depends on the baseline rate in addition to the log-rate effect. In the ZI case, τ also mixes the count-rate channel $\Delta\lambda$ and the ZI-probability channel Δp_{zi} from (7).

Why this parameterisation. Three properties of the construction are worth flagging. First, the model respects the non-negativity of Y automatically: $\lambda > 0$ and (where applicable) $p_{\text{zi},\, z} \in (0, 1)$ regardless of the forest values, so no truncation or clipping is required. Second, all forests are parameterised a priori independently, which is what permits the BCF differential-shrinkage idea [Hahn et al., 2020, §2.3] to be applied per group: each $\tau^{(g)}$ can be shrunk harder than its $\mu^{(g)}$ counterpart without disturbing other groups' priors. Third, the construction reduces to log-linear BART [Murray, 2021] when $\tau^{(g)} \equiv 0$ for all g, and to standard BCF in the limit where τ_f is shrunk hard to zero. In this sense CountBCF is the smallest extension of log-linear BART that delivers a separate moderating forest on each channel, and the smallest extension of BCF that handles count and zero-inflated count outcomes natively. The redundant (f_0, f_1) parameterisation in the ZI case is retained for the reasons given in Section 4.5: conjugate leaf updates, well-defined posterior on the identified functionals, and substantially better MCMC mixing [Murray, 2021, Supplement §A]. The BCF (μ, τ) decomposition does not break the redundancy: shifting μ_{f_0} and μ_{f_1} by a common constant leaves η_{zi} invariant, and shifting τ_{f_0} and τ_{f_1} by a common constant leaves $\Delta\eta_{\mathrm{zi}}$ invariant.

5.2 Priors

The CountBCF prior factorises across forests, the dispersion parameter, and the GIG-mixture latent structure of each leaf. The construction is symmetric across function groups, so we state the prior for a generic forest pair $(\mu^{(g)}, \tau^{(g)})$ and specialise the concentration hyperparameters to the count and ZI groups.

Sum-of-trees priors on each forest. Each forest is a log-linear BART regression in the sense of Section 4.2: a sum of $m_{s,g}$ shallow trees with tree structures $\{T_{s,g,h}\}$ drawn a priori independently from the CGM tree-structure prior and leaf parameters $\{\Lambda_{s,g,h}\} = \{\lambda_{s,g,h,t}\}$ drawn a priori independently from the GIG mixture (14). Collecting these, for $s \in \{\mu, \tau\}$ and (in the ZI case) $g \in \{0, 1, 2\}$,

$$\boxed{\begin{array}{rclrcl} T_{s,g,h} \mid \alpha_s, \beta_s & \overset{\text{iid}}{\sim} & \mathrm{CGM}(\alpha_s, \beta_s), & & & h = 1, \ldots, m_{s,g}, \\ \lambda_{s,g,h,t} \mid c_{s,g}, d_{s,g} & \overset{\text{iid}}{\sim} & \frac{1}{2}\,\mathrm{GIG}(-c_{s,g}, 2d_{s,g}, 0) + \frac{1}{2}\,\mathrm{GIG}(c_{s,g}, 0, 2d_{s,g}), & & & t = 1, \ldots, b_{s,g,h}, \\ \mu^{(g)}(x, \widehat{\pi}(x)) & = & \displaystyle\sum_{h=1}^{m_{\mu,g}} g\big((x, \widehat{\pi}(x));\, T_{\mu,g,h}, \Lambda_{\mu,g,h}\big), & \tau^{(g)}(x) & = & \displaystyle\sum_{h=1}^{m_{\tau,g}} g\big(x;\, T_{\tau,g,h}, \Lambda_{\tau,g,h}\big), \end{array}} \tag{24}$$

where the CGM density assigns an internal-node split probability $\alpha_s\,(1 + d)^{-\beta_s}$ at depth d and a uniform draw over splitting variable and threshold given the split. The depth-penalty hyperparameters (α_s, β_s) are shared across function groups (so all prognostic forests have the same tree-shape prior, and similarly for the moderating forests); the leaf concentration is allowed to differ across g to reflect the different natural scales of the count log-rate versus the zero-inflation log-odds. In the count-only regime, indices $g \in \{1, 2\}$ are dropped throughout.

Differential shrinkage applied symmetrically. CountBCF inherits the BCF [Hahn et al., 2020] idea that the treatment-effect surface $\tau^{(g)}$ should be shrunk more aggressively than the prognostic surface $\mu^{(g)}$ on each channel: the signal-to-noise in $\tau^{(g)}$ is typically lower than in $\mu^{(g)}$; spurious heterogeneity in any of $\tau_f, \tau_{f_0}, \tau_{f_1}$ manifests as fake interactions with Z on the respective channel; and a stub-biased prior on every $\tau^{(g)}$ delivers a near-constant treatment-effect estimator as a default. The default hyperparameters are summarised in Table 2. The tree-prior $(\alpha_\tau, \beta_\tau) = (0.25, 3)$ sharply downweights non-stub trees in every moderating forest: under it the prior probability of a non-trivial split at the root is 0.25 and the probability of a depth-2 internal node is approximately 0.0039, so on average each moderating forest is close to a piecewise-constant surface with very few splits. The leaf concentration $a_{0,\tau} = a_0/2$ (count group) and $z_{\mathrm{conc},\,\tau} = z_{\mathrm{conc}}/2$ (ZI groups) further shrinks the amplitude of those few splits: under the GIG-mixture calibration of Section 4.3, $\mathrm{Var}(\log \lambda_{\tau,g,h,t}) \approx a^2_{0,\tau,g}/m_{\tau,g}$, so the implied marginal prior on $\tau^{(g)}(x)$ is $\mathrm{N}(0, a^2_{0,\tau,g})$, a factor of two tighter than the prior on $\mu^{(g)}$.

The user-facing concentration a_0 defaults to $\frac{1}{2}(\log y_{(0.95)} - \overline{\log \mu_{0,i}})$, where $y_{(0.95)}$ is the empirical 95% quantile of y, centring the count-channel prior on a data-matched scale. The ZI concentration

Table 2: CountBCF default hyperparameters. The ZI rows ($g \in \{1, 2\}$) apply only when `count_model` $\in$ {`zipoisson`, `zinb`}; the dispersion parameters apply only when `count_model` $\in$ {`nb`, `zinb`}.

Hyperparameter	Default	Role
$m_{\mu,0}$, $m_{\tau,0}$	250, 50	Trees per forest, count group $g = 0$
$m_{\mu,1}$, $m_{\tau,1}$	100, 50	Trees per forest, ZI group $g = 1$ (f_0)
$m_{\mu,2}$, $m_{\tau,2}$	100, 50	Trees per forest, ZI group $g = 2$ (f_1)
(α_μ, β_μ)	(0.95, 2)	CGM depth penalty, all $\mu^{(g)}$
$(\alpha_\tau, \beta_\tau)$	(0.25, 3)	CGM depth penalty, all $\tau^{(g)}$
a_0	$\frac{1}{2}(\log y_{(0.95)} - \overline{\log \mu_{0,i}})$	Leaf concentration, μ_f (count)
$a_{0,\tau}$	$a_0/2$	Leaf concentration, τ_f (count)
z_{conc}	$3.5/\sqrt{2}$	Leaf concentration, μ_{f_0}, μ_{f_1} (ZI)
$z_{\text{conc},\tau}$	$z_{\text{conc}}/2$	Leaf concentration, τ_{f_0}, τ_{f_1} (ZI)
(κ_a, κ_b)	(5, 3)	Beta-prime prior on κ
σ_κ	0.21	RW-MH proposal SD on $\log \kappa$
`include_pihat`	`"control"`	Propensity appended to all $\mu^{(g)}$

z_{conc} defaults to $3.5/\sqrt{2} \approx 2.47$, matching the calibration of Murray [2021, §4.2.2] for the redundant (f_0, f_1) parameterisation. Both can be overridden by the user.

Propensity-as-covariate fix on every channel. A central insight of Hahn et al. [2020] is that, on the additive response scale, naively regressing the outcome on (X, Z) with a BART-like prior leaves the estimator vulnerable to RIC: when the propensity score $\pi(x)$ is a function of x, the prior shrinkage of the prognostic surface toward zero biases τ_f in a direction that depends on π. The fix is to append $\widehat{\pi}(x)$ to the prognostic forest's covariates so that μ_f can "absorb" the part of the outcome explained by π without recruiting τ_f. On the log-rate scale of (18) the RIC argument is at least as severe: a perturbation δ in μ_f translates to a multiplicative perturbation $e^\delta - 1 \approx \delta$ in the baseline rate, which then multiplies into τ via (23); a bias in μ_f correlated with π propagates linearly into τ. In the ZI case the same RIC argument applies independently on each channel: perturbations in μ_{f_0} or μ_{f_1} propagate into the ZI-channel moderating forests through the redundant parameterisation. We therefore append $\widehat{\pi}$ to the covariate set of every prognostic forest by default and do not append it to any moderating forest. The R wrapper exposes this as `include_pihat = "control"`, with `"moderate"`, `"both"`, and `"none"` available for sensitivity checks.

Negative-binomial dispersion κ. Under the NB / ZINB likelihood, the size parameter κ controls overdispersion: $\mathrm{Var}(Y_i \mid X_i, Z_i, S_i = 0) = \lambda_i + \lambda_i^2/\kappa$. Following Murray [2021] we place a Beta-prime prior on κ implied by

$$\frac{\kappa}{1+\kappa} \sim \mathrm{Beta}(\kappa_a, \kappa_b), \qquad (\kappa_a, \kappa_b) = (5,\ 3), \tag{25}$$

with mean 2.5 and a right tail heavy enough not to rule out near-Poisson behaviour while still penalising arbitrarily large overdispersion. The same prior governs κ under both NB and ZINB; structural zeros enter the κ-conditional likelihood only through z_i (Section 5.3). The dispersion is updated by a random-walk MH step on $\log \kappa$ with Gaussian proposal of standard deviation $\sigma_\kappa = 0.21$.

Table 3: Per-group sufficient statistics. The values u_i, v_i are inserted into (27) through the leaf accumulators $r_{s,g,h,t}, s_{s,g,h,t}$. The BCF weight is $\omega_i = 1$ for prognostic forests ($s = \mu$) and $\omega_i = Z_i$ for moderating forests ($s = \tau$); it premultiplies (u_i, v_i) before the leaf sum is formed. The non-ZI sub-model retains only the $g = 0$ row.

g	Function target	$u_i \cdot \omega_i^{-1}$	$v_i \cdot \omega_i^{-1}$
0	count rate λ	$z_i\, y_i$	$z_i\, \exp(\log \xi_i + \log \lambda(\mathbf{X}_i, Z_i) - \omega_i \log \lambda_{s,g,h,t})$
1	ZI log-odds f_0	$1 - z_i$	$\exp(\log \phi_i + \log f_0(\mathbf{X}_i) - \omega_i \log \lambda_{s,g,h,t})$
2	ZI log-odds f_1	z_i	$\exp(\log \phi_i + \log f_1(\mathbf{X}_i) - \omega_i \log \lambda_{s,g,h,t})$

5.3 Posterior computation

Posterior inference for CountBCF uses a Gibbs sampler that interleaves backfitting updates of the forests with latent-variable augmentation steps and (under NB / ZINB) a MH step on the dispersion. The structure mirrors the log-linear BART sampler of Murray [2021, §3.3, §S.3]: the only forest-level change is the introduction of a per-observation weight $\omega_i \in \{1, Z_i\}$ that routes treated and control units to the moderating and prognostic forests, replicated across all function groups.

Latent-variable augmentation. The augmentations are inherited from Sections 4.4 and 4.5. Each observation carries up to three latents:

$$\begin{aligned}
z_i \;\mid\; - \;&\sim\; \text{Bernoulli}\Big(\frac{(1-\omega(\mathbf{X}_i))\, p_{\text{count}}(0 \mid \mathbf{X}_i)}{\omega(\mathbf{X}_i) + (1-\omega(\mathbf{X}_i))\, p_{\text{count}}(0 \mid \mathbf{X}_i)}\Big) \quad \text{when } Y_i = 0 \text{ (ZIP/ZINB)}; \;\; z_i \equiv 1 \text{ when } Y_i > 0,\\
\xi_i \;\mid\; - \;&\sim\; \begin{cases} \delta_1 & \text{(Poisson / ZIP)},\\ \text{Gamma}\big(\kappa + Y_i,\; \kappa + \lambda(\mathbf{X}_i, Z_i)\big) & \text{(NB / ZINB)}, \end{cases}\\
\phi_i \;\mid\; - \;&\sim\; \text{Exponential}\big(\exp(f_0(\mathbf{X}_i)) + \exp(f_1(\mathbf{X}_i))\big) \quad \text{(ZIP/ZINB only)}.
\end{aligned} \tag{26}$$

Under the non-ZI sub-models the z_i and ϕ_i updates are skipped; under the Poisson / ZIP sub-models the ξ_i update is skipped.

Tree updates: a single primitive applied to all forests. Conditional on the latents, the joint conditional likelihood of the forests factorises into independent terms, each of which has the conditionally Gamma form (11). For forest s in function group g at tree h, the conditional augmented likelihood for the $b_{s,g,h}$ leaves of $T_{s,g,h}$ is

$$\boxed{L\big(\Lambda_{s,g,h} \,\big|\, T_{s,g,h}, T_{-(s,g,h)}, \Lambda_{-(s,g,h)}, z_i, \xi_i, \phi_i, Y, \kappa\big) \;\propto\; \prod_{t=1}^{b_{s,g,h}} \lambda_{s,g,h,t}^{r_{s,g,h,t}} \; \exp\big[-s_{s,g,h,t}\, \lambda_{s,g,h,t}\big],} \tag{27}$$

where the leaf sufficient statistics $r_{s,g,h,t} = \sum_{i \in \mathcal{A}_{s,g,h,t}} u_i$ and $s_{s,g,h,t} = \sum_{i \in \mathcal{A}_{s,g,h,t}} v_i\, f^{-(s,g,h)}(\mathbf{X}_i)$ aggregate over the units in the partition cell, using the per-group (u_i, v_i) enumerated in Table 3. The BCF weight $\omega_i \in \{1, Z_i\}$ enters multiplicatively in both u_i and v_i: for prognostic forests every unit contributes; for moderating forests only treated units contribute. This is the BCF moderating-forest ingredient transposed to the count (or ZI count) likelihood, applied symmetrically to all function groups present in the model.

The GIG-mixture leaf prior (14) is conjugate to (27), so the leaf draw is a closed-form mixture of GIG variates (implemented in `drmu_loglinear`), and $T_{s,g,h}$ is updated by a MH step with grow / prune / change / swap proposals (implemented in `bd_loglinear`). The MH acceptance ratio uses the integrated marginal likelihood obtained by integrating $\Lambda_{s,g,h}$ against the GIG-mixture prior; because both prior and likelihood are products over leaves, that integral factorises and reduces to a ratio of GIG normalising constants [Murray, 2021, Appendix A].

Per-group log-fit accumulator. Implemented as backfitting, the tree update maintains running accumulators on the n-vector index k: `allfit`$[k]$ tracks the total log-mean / log-odds; `allfits`$[s][k]$ tracks the contribution of forest s; and the per-group accumulator `log_f_per_group`$[g][k] = \mu^{(g)}(\mathbf{X}_k, \widehat{\pi}(\mathbf{X}_k)) + Z_k\, \tau^{(g)}(\mathbf{X}_k)$ stores the in-sample log-fit of both forests in group g. Before sampling tree (s, g, h) its current contribution is subtracted from all relevant accumulators; sufficient statistics are computed from the partial fit; $T_{s,g,h}$ and $\Lambda_{s,g,h}$ are updated; and the new contribution is added back. The per-group accumulator makes the six-forest update efficient: the ZI sufficient statistics read $\log f_0$ and $\log f_1$ from `log_f_per_group`$[1]$ and `log_f_per_group`$[2]$ in $O(1)$ per observation. After each sweep the soft mixture weight that enters the z_i update is recomputed as $\log w(\mathbf{X}_i) = \log f_1(\mathbf{X}_i) - \operatorname{logsumexp}(\log f_0(\mathbf{X}_i), \log f_1(\mathbf{X}_i))$, which avoids the underflow of naive computation when one of the log-fits is large and negative.

Dispersion update (NB / ZINB only). Propose $\kappa^\star = \exp(\log\kappa + \sigma_\kappa\, \varepsilon)$ with $\varepsilon \sim \mathrm{N}(0, 1)$. Accept with probability

$$\alpha_\kappa \;=\; \min\left\{1,\; \frac{p_{\mathrm{NB}}(Y \mid z_i, \lambda, \kappa^\star)}{p_{\mathrm{NB}}(Y \mid z_i, \lambda, \kappa)} \cdot \frac{p(\kappa^\star)}{p(\kappa)} \cdot \frac{\kappa^\star}{\kappa}\right\}, \tag{28}$$

where the likelihood factor is the augmented NB likelihood conditional on the mixture indicator (so structural zeros do not enter the dispersion update under ZINB), $p(\kappa)$ is the Beta-prime density implied by (25), and the Jacobian factor $\kappa^\star/\kappa$ comes from the log-scale proposal. Acceptance rates after burn-in are returned as `kappa_acpt` and serve as a coarse convergence diagnostic; we target the 0.20–0.45 band typical for random-walk MH.

Algorithm 1 CountBCF: one MCMC sweep. Indices $s \in \{\mu, \tau\}$ range over forests, $g \in \{0\}$ in the count sub-model and $g \in \{0, 1, 2\}$ in the ZI sub-model, and $h = 1, \ldots, m_{s,g}$ over the trees of forest (s, g). The BCF weight is $\omega_i = 1$ for $s = \mu$ and $\omega_i = Z_i$ for $s = \tau$. Latent updates that are not applicable to the active sub-model are skipped.

Require: Current state $\big(\{T_{s,g,h}, \Lambda_{s,g,h}\},\ \kappa,\ \{z_i, \xi_i, \phi_i\}\big)$.

1: **Dispersion update (NB / ZINB only).** Propose $\kappa^\star = \exp(\log\kappa + \sigma_\kappa\,\varepsilon)$, $\varepsilon \sim \mathrm{N}(0, 1)$. Accept with probability α_κ from (28); otherwise retain κ.

2: **Augmentation update for z_i (ZIP / ZINB only).** For each i with $Y_i = 0$, draw z_i from the Bernoulli full conditional in (26); set $z_i = 1$ when $Y_i > 0$. [`drz_loglinear`]

3: **Augmentation update for ξ_i (NB / ZINB only).** For each i, draw ξ_i from the Gamma full conditional; set $\xi_i \equiv 1$ otherwise. [`drxi_loglinear`]

4: **Augmentation update for ϕ_i (ZIP / ZINB only).** For each i, draw $\phi_i \sim \mathrm{Exponential}(\exp(f_0(\mathbf{X}_i)) + \exp(f_1(\mathbf{X}_i)))$. [`drphi_loglinear`]

5: **Backfitting tree updates.** For each function group g active in the sub-model, each forest $s \in \{\mu, \tau\}$, and each tree $h = 1, \ldots, m_{s,g}$:

5.1 Subtract the current contribution of $(T_{s,g,h}, \Lambda_{s,g,h})$ from `allfit`, `allfits`$[s]$, and `log_f_per_group`$[g]$.

5.2 Compute the leaf sufficient statistics $(r_{s,g,h,t}, s_{s,g,h,t})$ from (27) using the per-group (u_i, v_i) of Table 3, premultiplied by the weight ω_i.

5.3 Propose a tree move (grow / prune / change / swap) for $T_{s,g,h}$ and accept with probability from the integrated marginal likelihood (GIG-mixture conjugacy). [`bd_loglinear`]

5.4 Draw the leaf values $\Lambda_{s,g,h} \mid T_{s,g,h}, z_i, \xi_i, \phi_i, Y, \kappa$ from the GIG-mixture posterior implied by (27). [`drmu_loglinear`]

5.5 Add the new contribution back into the affected accumulators.

6: **Soft mixture weight (ZIP / ZINB only).** For each i, recompute $\log w(\mathbf{X}_i) = \log f_1(\mathbf{X}_i) - \mathrm{logsumexp}(\log f_0(\mathbf{X}_i), \log f_1(\mathbf{X}_i))$.

7: **(Optional) Random-effects update.** If enabled, draw the random-intercept block from its Gibbs full conditional; otherwise skip.

8: **return** Updated state $\big(\{T_{s,g,h}, \Lambda_{s,g,h}\},\ \kappa,\ \{z_i, \xi_i, \phi_i\}\big)$.

Implementation lineage. The low-level C++ primitives and dispersion Metropolis-Hastings steps are inherited verbatim from the `countbart` package[1]; the CountBCF contribution is the custom dispatch logic[2] that (i) instantiates two forests per function group rather than one, (ii) sets the per-forest weight vector ω_i to either $\mathbf{1}_n$ or Z_{trt}, (iii) applies ω_i to the sufficient-statistic

[1]Specifically, the C++ kernels `bd_loglinear`, `drmu_loglinear`, `fit_loglinear`, `drz_loglinear`, `drxi_loglinear`, and `drphi_loglinear` of Wikle and Zigler [2023].

[2]Implemented in `src/countbcf.cpp`.

construction in step 5.2, and (iv) maintains the per-group log-fit accumulator so that the four ZI forests read f_0 and f_1 in $O(1)$ per observation. The change is mechanical at the kernel level but yields a sampler that, to our knowledge, is functionally distinct from any log-linear BART variant in the published literature: it jointly estimates up to six log-rate / log-odds surfaces under differential shrinkage, with each moderating forest seeing only the treated subsample's residual variation.

5.4 Posterior summaries

Algorithm 1 produces a sequence of B post-burn-in posterior draws indexed by $b = 1, \dots, B$. The sampler stores, for each draw, the per-unit values of the active forests on the original (unsorted) input order:

$$\begin{aligned} \mu_f^{(b)}(X_i) &= \texttt{mu\_f\_post}[b,i], & \tau_f^{(b)}(X_i) &= \texttt{tau\_f\_post}[b,i], \\ \mu_{f_0}^{(b)}(X_i) &= \texttt{mu\_f0\_post}[b,i], & \tau_{f_0}^{(b)}(X_i) &= \texttt{tau\_f0\_post}[b,i], \\ \mu_{f_1}^{(b)}(X_i) &= \texttt{mu\_f1\_post}[b,i], & \tau_{f_1}^{(b)}(X_i) &= \texttt{tau\_f1\_post}[b,i], \end{aligned} \tag{29}$$

together with the dispersion draws $\kappa^{(b)}$ (NB / ZINB only). In the non-ZI sub-model only the first row is stored. Downstream inference is a post-processing exercise on these matrices and produces three (count) or four (ZI) classes of summaries.

Per-iteration potential outcomes. For each draw b and each unit i, the closed-form mapping from stored forest values to response-scale potential-outcome means is, for $z \in \{0, 1\}$,

$$\boxed{\begin{aligned} \log \lambda_z^{(b)}(X_i) &= \mu_f^{(b)}(X_i) + z \cdot \tau_f^{(b)}(X_i), \\ \eta_{\text{zi}}[z]^{(b)}(X_i) &= \big[\mu_{f_0}^{(b)}(X_i) + z \cdot \tau_{f_0}^{(b)}(X_i)\big] - \big[\mu_{f_1}^{(b)}(X_i) + z \cdot \tau_{f_1}^{(b)}(X_i)\big], \\ \mu_z^{(b)}(X_i) &= \big(1 - \sigma(\eta_{\text{zi}}[z]^{(b)}(X_i))\big) \exp\!\big(\log \lambda_z^{(b)}(X_i)\big). \end{aligned}} \tag{30}$$

In the non-ZI sub-model $\sigma(\eta_{\text{zi}}) \equiv 0$ and (30) collapses to $\mu_z^{(b)}(X_i) = \exp(\mu_f^{(b)}(X_i) + z\tau_f^{(b)}(X_i))$. The pair $(\mu_0^{(b)}, \mu_1^{(b)})$ collects the response-scale potential-outcome means at every unit under every draw, and all reported quantities below are deterministic functions of these two matrices – no additional sampling is required: posterior uncertainty in the response-scale CATE arises from joint uncertainty in the forests and is automatically propagated by the backfitting Gibbs sampler.

Response-scale CATE. The primary estimand is (6). The bth posterior draw is

$$\widehat{\tau_{,i}}^{(b)} \;=\; \mu_1^{(b)}(X_i) - \mu_0^{(b)}(X_i), \tag{31}$$

with the count-only form $\widehat{\tau_{,i}}^{(b)} = \exp(\mu_f^{(b)}(X_i))(\exp(\tau_f^{(b)}(X_i)) - 1)$ recovered in the non-ZI sub-model. The per-unit posterior mean and 95% equal-tailed credible interval are $\overline{\tau_{,i}} = B^{-1}\sum_b \widehat{\tau_{,i}}^{(b)}$ and $[Q_{0.025}, Q_{0.975}]$ across draws, with Q_q the empirical q-quantile. The log-rate CATE $\widehat{\tau_{f,i}}^{(b)} = \tau_f^{(b)}(X_i)$ is reported alongside for users whose applied questions are best phrased as relative rates.

Sample-average treatment effect. The ATE (6) is the in-sample mean of the response-scale CATE,

$$\widehat{\text{ATE}}^{(b)} \;=\; \frac{1}{n}\sum_{i=1}^{n} \widehat{\tau_{,i}}^{(b)}, \tag{32}$$

giving a B-vector of posterior draws. The posterior mean and 95% equal-tailed interval of this vector give the point estimate and uncertainty of the sample ATE. The average is over the empirical distribution of X, so (32) estimates the sample-average treatment effect rather than a population-average one; Hahn et al. [2020] discuss why this is the natural target for BART-family methods.

Channel decomposition (ZI only). A central affordance of the six-forest construction over a single-channel causal forest is that τ can be decomposed into a count-rate channel and a structural-zero channel. The per-unit posterior draws of the two channel components introduced in (7) are

$$\widehat{\Delta\lambda_{,i}}^{(b)} \;=\; \exp\!\big(\log\lambda_1^{(b)}(X_i)\big) - \exp\!\big(\log\lambda_0^{(b)}(X_i)\big), \qquad \widehat{\Delta p_{\mathrm{zi},i}}^{(b)} \;=\; \sigma\!\big(\eta_{\mathrm{zi}}[1]^{(b)}(X_i)\big) - \sigma\!\big(\eta_{\mathrm{zi}}[0]^{(b)}(X_i)\big), \tag{33}$$

both deterministic functions of the stored forest values. Either channel can be summarised by its posterior mean and 95% credible interval per unit, or averaged to produce a posterior of the channel-specific sample average. The two channels are not constrained to share a sign: for some units the treatment can increase the count rate and decrease the structural-zero probability (both contributing positively to τ); for other units the channels can offset, producing a small response-scale effect masking large channel-specific shifts.

6 Research methodology

This section pre-registers the simulation study that will evaluate CountBCF against the natural Gaussian-likelihood baseline. The specification is held fixed before any cells are run; results will be reported in Section 7 without revision to the design below. The motivation for pre-registration is standard [Hahn et al., 2020, § 3.2]: heterogeneous-effect comparisons are sensitive to which data-generating processes (DGPs), factor sweeps, and metrics one chooses to highlight, and reporting those choices after seeing results risks confirming the preferred method.

6.1 Estimand

We target the response-scale conditional and average treatment effects defined in Section 3: for unit i, the potential-outcome means $\mu_z(x) = (1 - p_{\mathrm{zi},\,z}(x))\exp(\log\lambda_z(x))$ of (5) give rise to

$$\tau(x) \;=\; \mu_1(x) - \mu_0(x), \qquad \mathrm{ATE} \;=\; \mathbb{E}_X\big[\tau(X)\big], \tag{34}$$

on the same units as Y. Under zero-inflation (34) mixes a change in the count rate and a change in the structural-zero probability, and a method that changes only one of the two will systematically under-estimate the effect on at least some part of the covariate space; the ZI DGPs of Section 6.2 engineer this asymmetry deliberately. All methods, regardless of their native scale, are scored against this rate-scale truth. For CountBCF the per-draw CATE is obtained from (30)–(31); for the Gaussian BCF baseline it is read off directly from the moderating forest.

6.2 Data-generating processes

The simulation grid is a 2×4 design over (i) the functional form of the prognostic and moderating surfaces (linear or nonlinear) and (ii) the count distribution (Poisson, NB with $\kappa = 2$, ZIP, or ZINB with $\kappa = 2$). All eight DGPs share the same covariate distribution and propensity model, so cross-DGP comparisons isolate the effects of nonlinearity, overdispersion, and zero-inflation. Each DGP features strong, sign-changing, multi-covariate heterogeneity; the ZI DGPs additionally contain non-trivial heterogeneity on the structural-zero channel, so the per-unit CATE estimation task is substantively challenging in every cell.

Covariates and treatment. We draw n iid covariate vectors $X_i \in \mathbb{R}^p$ with the first $p-2$ columns iid standard Gaussian and the last two binary,

$$X_{i,j} \overset{\text{iid}}{\sim} \mathrm{N}(0,1) \text{ for } j = 1,\dots,p-2, \quad X_{i,\,p-1} \sim \text{Bernoulli}(0.6), \quad X_{i,\,p} \sim \text{Bernoulli}(0.4), \tag{35}$$

so every design contains two binary columns regardless of p. Only the first five covariates carry signal; columns $X_{\cdot,6},\dots,X_{\cdot,p-2}$ are irrelevant noise included to stress-test variable selection as p grows. The propensity model is fixed across DGPs at

$$\pi(x) \;=\; \sigma(0.5\,x_1 - 0.4\,x_2), \qquad Z_i \mid X_i \sim \text{Bernoulli}(\pi(X_i)), \tag{36}$$

with $\sigma(\cdot)$ the logistic CDF. The estimated propensity $\widehat{\pi}$ passed to the methods is set to the truth, $\widehat{\pi}(X_i) = \pi(X_i)$, so the comparison isolates the outcome model rather than penalising methods for propensity mis-estimation.

Count component. For all eight DGPs the conditional log-mean of the count component (non-ZI cells) or the not-structural-zero component (ZI cells) has the BCF-style form

$$\log \lambda_i \;=\; \mu_c(X_i) \;+\; Z_i \left[\tau_0 + \tau_h(X_i)\right], \tag{37}$$

with the prognostic surface μ_c and the heterogeneous part τ_h taking the form in the upper half of Table 4. The scalar τ_0 is a per-cell intercept calibrated so that the realised response-scale ATE matches the cell's target value. For non-ZI cells, we solve

$$\mathbb{E}_X\!\left[\exp(\mu_c(X))\left(\exp(\tau_0 + \tau_h(X)) - 1\right)\right] \;=\; \text{ATE}_{\text{target}} \tag{38}$$

via `R`'s `uniroot()` on a fresh held-out sample of 5,000 covariate vectors. For ZI cells, the calibration accounts for the $(1 - p_{\text{zi},\,z})$ thinning of the count component; only τ_0 is calibrated, the ZI shift $\Delta\eta_{\text{zi}}$ is held fixed.

Zero-inflation component (ZI cells only). The structural-zero probability follows $p_{\text{zi},\,z}(x) = \sigma(\eta_{\text{zi}}(x) + z\Delta\eta_{\text{zi}}(x))$, with the control logit $\eta_{\text{zi}}(x)$ and treatment shift $\Delta\eta_{\text{zi}}(x)$ in the lower half of Table 4. The control logit is calibrated so the baseline structural-zero probability is approximately 25–30% across the covariate distribution – large enough to make the ZI channel matter but small enough not to swamp the count signal. The treatment shift $\Delta\eta_{\text{zi}}(x)$ is sign-changing across units: for some units the treatment increases the structural-zero probability while for others it decreases it, with the sign and magnitude depending on X. A single ATE-level summary of the ZI channel would average over the sign reversal and report a misleading null; the channel-decomposition plots described below are designed to recover this heterogeneity.

Sampling distribution. For each unit i, the response is drawn from one of four distributions. Under the non-ZI cells, $Y_i \sim \text{Poisson}(\lambda_i)$ or $\text{NB}(\kappa, \lambda_i)$ with $\kappa = 2$. Under the ZI cells, a Bernoulli structural-zero indicator $S_i \sim \text{Bernoulli}(p_{\text{zi},\,Z_i}(X_i))$ is drawn first; if $S_i = 1$ then $Y_i = 0$ deterministically; otherwise $Y_i \mid S_i = 0 \sim \text{Poisson}(\lambda_i)$ or $\text{NB}(\kappa, \lambda_i)$ with $\kappa = 2$. Combined with Table 4 this yields the eight CATE-focused DGPs (linear and nonlinear, crossed with Poisson, NB, ZIP, and ZINB).

Table 4: Functional forms for the prognostic surface μ_c and the heterogeneous component τ_h (count channel, top), and for the control logit η_{zi} and treatment shift $\Delta\eta_{\text{zi}}$ on the ZI channel (bottom). The linear and nonlinear families each cross with the four count distributions $\{\text{Poisson}, \text{NB}, \text{ZIP}, \text{ZINB}\}$ to give eight DGPs; the ZI rows apply only to the ZIP / ZINB cells.

DGP family	linear	nonlinear
Count component ($\log \lambda_i = \mu_c(X_i) + Z_i[\tau_0 + \tau_h(X_i)]$)		
$\mu_c(x)$	$1.0 + 0.5x_1 - 0.3x_2 + 0.2x_3$	$0.8 + 0.6\sin(x_1) + 0.4x_2x_3 + 0.5\,\mathbf{1}\{x_4 > 0\}$
$\tau_h(x)$	$0.60x_1 - 0.45x_2 + 0.30x_3$	$0.80\sin(x_1) + 0.50\,\mathbf{1}\{x_2 > 0\}\,x_3 - 0.40x_5$
ZI component ($p_{\text{zi},\,z}(x) = \sigma(\eta_{\text{zi}}(x) + z\Delta\eta_{\text{zi}}(x))$, ZIP / ZINB only)		
$\eta_{\text{zi}}(x)$	$-1.0 + 0.5x_2$	$-1.0 + 0.7\lvert x_2\rvert - 0.5\exp(-x_3^2)$
$\Delta\eta_{\text{zi}}(x)$	$-0.50x_1 + 0.30x_3 - 0.40x_4$	$-0.60x_1x_4 + 0.40x_2\,\mathbf{1}\{x_5 > 0\}$

Table 5: One-at-a-time factor sweep. "Held fixed" columns list the factors that take their reference value within the row's sweep.

Factor	Off-reference levels	Held fixed
Sample size N	{100, 500}	$p = 5$, ATE $= 1.25$
Covariate count p	{50, 250}	$N = 250$, ATE $= 1.25$
Target ATE	{0.5, 2.5}	$N = 250$, $p = 5$

6.3 Experimental factors

We sweep three experimental factors – sample size N, covariate count p, and target ATE – one-at-a-time (OAT) around a reference cell. Reference values are $N_{\text{ref}} = 250$, $p_{\text{ref}} = 5$, $\text{ATE}_{\text{ref}} = 1.25$. Off-reference levels are in Table 5; together with the reference, this gives $1 + 2 + 2 + 2 = 7$ cells per $(\text{method}, \text{DGP})$ pair. For each cell all eight DGPs of Section 6.2 are run. Within a cell $\times$ DGP we draw $N_{\text{sim}} = 100$ independent Monte Carlo replicates; the seed of replicate s is exactly s – `set.seed(sim_id)` – so any single replicate is reproducible from its $(\text{dgp}, N, p, \text{ATE}, \text{sim\_id})$ tuple alone.

6.4 Methods compared

Both methods produce a posterior over the response-scale ATE and per-unit posterior means and credible intervals for the response-scale CATE; both are Bayesian, both share the BCF prior structure, and they differ only in the outcome likelihood.

1. **CountBCF (proposed).** Fit via `countbcf()` with `count_model` set to one of `"poisson"`, `"nb"`, `"zipoisson"`, or `"zinb"` depending on the DGP. All other hyperparameters take the defaults catalogued in Table 2. The per-unit CATE is recovered from the closed-form mapping (30)–(31); under ZIP / ZINB the same coefficient matrices are also post-processed into the channel decomposition $(\widehat{\Delta\lambda}, \widehat{\Delta p_{\text{zi}}})$ of (33).

2. **BCF (Gaussian baseline).** The Hahn et al. [2020] BCF applied to the count outcome as if it were Gaussian, fit via `bcf_binary()`. The treatment-effect posterior is read off the moderating forest with `get_forest_fit()`. This is the natural "what if you ignored the

count structure" baseline: it retains the BCF (μ, τ) decomposition and the propensity-as-covariate fix but does not respect the discreteness of Y, the heteroscedasticity induced by the count likelihood, or the structural-zero process; in the ZI case it produces only a single response-scale CATE with no channel decomposition.

3. **Parametric Generalised Linear Models (Poisson/NB or ZIP/ZINB GLM).** In the count-only setting, we fit a Poisson GLM and a Negative-Binomial GLM with a log link, including all covariates, treatment, and treatment-by-covariate interactions. In the zero-inflated setting, we fit Zero-Inflated Poisson (ZIP) and Zero-Inflated Negative-Binomial (ZINB) GLMs via the `pscl` package [Zeileis et al., 2008], including treatment-by-covariate interactions in both the count and the zero-inflation components. To obtain posterior draws and uncertainty intervals for these models, we simulate coefficients from their multivariate normal asymptotic sampling distributions.

4. **Causal Forest.** Fit via the `grf` package [Athey et al., 2019]. This is a nonparametric frequentist random forest optimized for treatment effect estimation. The aggregate average treatment effect (ATE) and its standard errors are obtained using the doubly-robust augmented inverse probability weighting (AIPW) estimator, while the individual-level treatment effects (CATEs) are evaluated directly using the forest predictions.

Both Bayesian methods use the same MCMC budget: `nburn` = `nsim` = 1000, `nthin` = 1. The methodological symmetry is deliberate – the only structural difference is the outcome likelihood – so any performance gap observed in Section 7 can be attributed to the likelihood, not to incidental choices of prior, MCMC budget, or post-processing.

CountBART considered and dropped. An S-learner built on top of `count_bart()` – augmenting the design matrix with Z and $\widehat{\pi}$, fitting a single log-linear BART model on $(X, Z, \widehat{\pi})$ (three-forest in the ZI case), and reading the CATE off counterfactual evaluations – was originally part of the benchmark. It consistently exhausts the available Colab RAM and crashes on the larger ablation cells ($p \in \{50, 250\}$); the three-forest backbone of the ZI model amplifies the memory pressure relative to the non-ZI case. Until the memory footprint of the upstream `count_bart` is reduced, the CountBART S-learner is not feasible to evaluate against CountBCF on the same hardware, and we exclude it rather than report it on the subset of cells where it runs.

6.5 Metrics

All metrics that involve the truth are computed on the response scale, regardless of which scale a method natively reports. Aggregating across the $N_{\text{sim}} = 100$ replicates within a cell gives Monte Carlo means and standard errors. Two families of metrics are reported, both targeted at the calibration of a method's posterior rather than only its point estimate.

ATE-level metrics. Let $\widehat{\text{ATE}}^{(r)}$ denote the posterior-mean ATE on replicate r, $\widehat{\text{CI}}_{95}^{(r)}$ its 95% equal-tailed credible interval, and $\text{ATE}^{(r)} = n^{-1} \sum_i \tau(X_i)$ the realised true ATE on the same replicate. We report:

- the cross-replicate root mean squared error $\text{RMSE}(\widehat{\text{ATE}}) = \sqrt{N_{\text{sim}}^{-1} \sum_r (\widehat{\text{ATE}}^{(r)} - \text{ATE}^{(r)})^2}$;
- the empirical 95% coverage $\text{Cov}_{95}(\widehat{\text{ATE}}) = N_{\text{sim}}^{-1} \sum_r \mathbf{1}\{\text{ATE}^{(r)} \in \widehat{\text{CI}}_{95}^{(r)}\}$; and
- the average 95% credible-interval width $\overline{W_{95}(\widehat{\text{ATE}})}$.

CATE-level metrics. On each Monte Carlo replicate $r \in \{1, \ldots, N_{\text{sim}}\}$, let $\widehat{\tau}^{(r)}(X_i)$ denote the posterior mean of $\tau(X_i)$, $\widehat{\mathrm{CI}}_{95,i}^{(r)}$ its unit-level 95% credible interval, and $\tau(X_i) = \mu_1(X_i) - \mu_0(X_i)$ the true CATE. For each replicate, we compute:

- the replicate-specific precision in estimating heterogeneous effects:

$$\mathrm{PEHE}^{(r)} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(\widehat{\tau}^{(r)}(X_i) - \tau(X_i)\right)^2}; \tag{39}$$

- the replicate-specific per-unit empirical 95% coverage:

$$\mathrm{Cov}_{95}^{(r)}(\widehat{\tau}) = \frac{1}{n}\sum_{i=1}^{n}\mathbf{1}\left\{\tau(X_i) \in \widehat{\mathrm{CI}}_{95,i}^{(r)}\right\}; \quad \text{and} \tag{40}$$

- the replicate-specific average per-unit 95% credible-interval width:

$$\overline{W_{95}^{(r)}(\widehat{\tau})} = \frac{1}{n}\sum_{i=1}^{n} W_{95,i}^{(r)}(\widehat{\tau}), \tag{41}$$

  where $W_{95,i}^{(r)}(\widehat{\tau})$ is the width of the interval $\widehat{\mathrm{CI}}_{95,i}^{(r)}$.

We report the Monte Carlo mean and standard deviation of these replicate-specific summaries across the $N_{\text{sim}} = 100$ replicates, written as mean $\pm$ SD in all tables. PEHE is the headline point-estimate metric, but in the $N_{\text{sim}} = 100$ regime the credible-interval metrics — particularly the per-unit coverage and credible-interval width — are what most clearly expose miscalibration of the Gaussian baseline. Additional secondary diagnostics (such as fitting time, absolute CATE bias, and structural-zero rates) are available in the replication archive.

6.6 Outputs and reproducibility

Each (method, DGP, cell) tuple is run for $N_{\text{sim}} = 100$ Monte Carlo replicates whose results are streamed incrementally, so that a partial run can be resumed without losing earlier replicates. With 2 methods, 8 CATE-focused DGPs, and 7 cells per method×DGP pair, this gives 112 aggregated cells in total across the count-only and zero-inflated sub-studies. Replication material and the full per-cell numbers are released alongside the `countbcf` R package.

Output panels. Section 7 presents two families of panels. Reference-cell tables report the six metrics above on each of the eight DGPs at the reference cell, for both methods. Channel decomposition — the panel template specific to the ZI sub-study — plots, for CountBCF only, the per-unit posterior means of $\Delta\lambda$ and Δp_{zi} against each other and against the combined response-scale CATE τ on a per-DGP basis at the reference cell. Channels that share sign across most of the covariate space appear as a clustered cloud along the diagonal; sign-changing DGPs produce a cross-shaped scatter; either pattern is a visual confirmation that the six-forest decomposition recovers heterogeneity that single-channel methods cannot expose. The one-at-a-time factor sweeps from Table 5 are reported as ablation panels in Appendix A.

7 Simulation Studies

This section reports the outcome of the pre-registered protocol of Section 6, with no revision to the design. Two sub-studies are run in parallel: a count-only sub-study over the four CATE-focused Poisson and Negative-Binomial DGPs of Table 4, and a zero-inflated sub-study over the corresponding ZIP and ZINB cells. The two sub-studies isolate complementary phenomena. The count-only sub-study measures the cost of treating an integer outcome as Gaussian when no structural-zero channel is present; the zero-inflated sub-study additionally stresses the methods on a second treatment channel ($\Delta\eta_{\text{zi}}$) that the Gaussian baseline cannot represent at all. Because both sub-studies share the covariate distribution, propensity model (36), factor grid (Table 5), and Monte Carlo budget ($N_{\text{sim}} = 100$ replicates per cell), every contrast across the two sub-studies can be attributed to the presence or absence of zero-inflation in the DGP, not to incidental design choices.

The remainder of the section is organised as follows. The reference-cell head-to-head (§ 7.1) fixes the design at $(N, p, \text{ATE}) = (250, 5, 1.25)$ and reports both methods on all six metrics across all eight DGPs. We additionally conducted one-at-a-time ablation sweeps over each of the three experimental factors of Table 5 — sample size, covariate count, and target ATE — to confirm that the ordering of the two methods at the reference cell is robust across the design space; § 7.2 summarises the conclusion and the full panels are deferred to Appendix A for curious readers. The cross-cutting interpretation (§ A.4) and a list of caveats (§ A.5) close the section. The underlying per-cell numbers and the aggregate ATE-level curves along the same sweeps are provided in the replication archive that accompanies the `countbcf` R package; the headline numbers are quoted in the text.

7.1 Reference-cell head-to-head

We first fix the design at the reference cell $(N, p, \text{ATE}) = (250, 5, 1.25)$ and compare the two methods at the centre of the factor grid. Tables 6 and 7 report all six metrics for the four count-only and four zero-inflated DGPs respectively. RMSE(ATE) and ATE coverage are scalar Monte Carlo summaries; the remaining columns are reported as cross-replicate mean $\pm$ one standard deviation across the 100 replicates per cell.

Parametric efficiency under correct specification. When the parametric functional form is correctly specified and the covariate dimension is small ($p = 5$), the parametric GLMs are the most accurate models on point estimates. In the linear/Poisson DGP (Table 6), the Poisson GLM achieves the lowest PEHE (1.09 ± 0.43) and the lowest ATE RMSE (0.30) of any method, outperforming the nonparametric CountBCF (PEHE = 4.77, RMSE = 1.08) and Causal Forest (PEHE = 7.67, RMSE = 0.52). A similar pattern is observed in the zero-inflated settings (Table 7), where the ZIP GLM leads on linear/ZIP (PEHE = 1.44 ± 0.89, RMSE = 0.35). This efficiency is a natural consequence of parametric asymptotics: when $p \ll N$ and the likelihood is correctly specified, parametric estimators converge at a rate of $O(1/\sqrt{N})$, which nonparametric models with wider prior support cannot match.

However, this efficiency comes at a cost in calibration. The Poisson GLM covers the ATE only 63% of the time and individual CATEs only 63% of the time in the count-only setting, and the ZIP GLM covers the ATE and CATE only 81% and 80% of the time, respectively. The Negative Binomial and ZINB GLMs, which carry an extra dispersion parameter, exhibit better calibration (NB GLM ATE coverage 0.89–0.92, ZINB GLM 0.94–0.96), but as shown in the ablation sweeps, they collapse when dimensionality increases.

Table 6: Reference-cell ($N = 250$, $p = 5$, ATE $= 1.25$) Monte-Carlo summary on the count-only DGPs (Poisson, NB). RMSE ATE and ATE coverage are scalar Monte-Carlo summaries; the remaining columns report the cross-replicate mean ± one standard deviation. ATE CI Width is the average width of the 95% ATE posterior credible interval; PEHE is $\sqrt{(\hat{\tau}_i - \tau_i)^2}$; CATE Cov. 95 is the per-unit 95% credible-interval coverage of $\tau(X_i)$; CATE CI Width95 is the mean width of those unit-level intervals.

		ATE			CATE		
DGP	Method	RMSE	Cov. 95	Avg. CI Width	PEHE	Cov. 95	Avg. CI Width95
Linear / Poisson	CountBCF	1.08	0.65	2.28 ± 0.82	4.77 ± 2.46	0.92 ± 0.07	7.83 ± 2.21
	BCF (Gaussian)	1.15	0.55	1.23 ± 0.26	6.23 ± 4.44	0.81 ± 0.08	4.20 ± 1.12
	Poisson GLM	**0.30**	0.89	**1.05** ± 0.10	**1.09** ± 0.43	**0.95** ± 0.07	**2.53** ± 0.18
	NB GLM	**0.30**	0.89	**1.05** ± 0.10	**1.09** ± 0.43	**0.95** ± 0.07	**2.53** ± 0.18
	Causal Forest	0.52	**0.99**	3.07 ± 1.21	7.67 ± 3.57	0.62 ± 0.15	4.46 ± 1.45
Nonlinear / Poisson	CountBCF	0.76	0.80	1.88 ± 0.44	**3.48** ± 1.66	**0.87** ± 0.09	6.74 ± 1.75
	BCF (Gaussian)	1.09	0.64	1.40 ± 0.18	5.92 ± 4.80	0.81 ± 0.07	5.32 ± 0.98
	Poisson GLM	0.62	0.67	**1.09** ± 0.08	4.76 ± 1.94	0.48 ± 0.12	**2.59** ± 0.15
	NB GLM	0.98	0.89	2.09 ± 0.57	5.90 ± 3.21	0.75 ± 0.10	4.67 ± 1.05
	Causal Forest	**0.61**	**0.96**	2.80 ± 1.24	5.33 ± 2.44	0.66 ± 0.12	4.04 ± 1.51
Linear / NB	CountBCF	1.25	0.65	2.95 ± 1.18	5.95 ± 2.83	0.90 ± 0.07	10.08 ± 3.09
	BCF (Gaussian)	1.21	0.60	1.79 ± 0.36	9.16 ± 4.90	0.82 ± 0.06	5.54 ± 1.63
	Poisson GLM	0.72	0.51	**1.04** ± 0.11	3.69 ± 2.33	0.68 ± 0.14	**2.54** ± 0.23
	NB GLM	**0.70**	**0.92**	2.53 ± 0.55	**2.98** ± 1.76	**0.93** ± 0.08	5.34 ± 0.86
	Causal Forest	0.86	**0.98**	3.97 ± 1.76	7.74 ± 3.56	0.72 ± 0.14	5.36 ± 2.09
Nonlinear / NB	CountBCF	0.90	0.80	2.63 ± 0.65	**4.49** ± 1.81	**0.88** ± 0.07	9.61 ± 2.95
	BCF (Gaussian)	1.30	0.68	2.23 ± 0.57	7.81 ± 5.95	0.83 ± 0.08	7.34 ± 2.07
	Poisson GLM	**0.87**	0.46	**1.08** ± 0.10	5.26 ± 2.03	0.42 ± 0.11	**2.59** ± 0.22
	NB GLM	1.17	0.88	2.98 ± 0.81	6.36 ± 3.12	0.79 ± 0.10	6.43 ± 1.53
	Causal Forest	**0.87**	**0.94**	3.53 ± 1.69	5.46 ± 2.41	0.71 ± 0.13	4.93 ± 2.28

Aggregate ATE vs. CATE calibration. The nonparametric Causal Forest is a strong, robust ATE estimator, maintaining near-nominal ATE coverage (ranging from 0.94 to 0.98) across both studies. However, its per-unit CATE coverage is severely limited, averaging only 68% in the count-only reference cell and 72% in the zero-inflated reference cell. This under-coverage occurs because the Causal Forest shrinks treatment effect heterogeneity too aggressively, resulting in unit-level credible intervals that are too narrow to cover the true $\tau(X_i)$.

Conversely, the Gaussian BCF baseline suffers from systematic under-coverage on both metrics: its average ATE coverage is 0.62 (count) and 0.66 (ZI), and its per-unit CATE coverage is 0.82 (count) and 0.83 (ZI). By approximating the integer-valued response with a homoscedastic Gaussian likelihood, the Gaussian BCF underestimates the posterior uncertainty, yielding narrow but highly miscalibrated intervals.

CountBCF performance and CATE estimation. CountBCF successfully navigates this trade-off. While it yields higher PEHE and RMSE than correctly specified parametric models in low-dimensional, linear settings, it remains the best nonparametric model. More importantly, CountBCF achieves excellent calibration for both ATE and CATE. In the zero-inflated study, CountBCF's per-unit CATE coverage reaches 0.93–0.94 at the reference cell, essentially at the nominal 0.95, while maintaining a low PEHE (e.g., 3.06 ± 1.47 in nonlinear/ZIP and 3.49 ± 1.27 in nonlinear/ZINB, outperforming both Gaussian BCF and Causal Forest).

Table 7: Reference-cell ($N = 250$, $p = 5$, ATE $= 1.25$) Monte-Carlo summary on the zero-inflated DGPs (ZIP, ZINB). Conventions match Table 6.

		ATE			CATE		
DGP	Method	RMSE	Cov. 95	Avg. CI Width	PEHE	Cov. 95	Avg. CI Width95
Linear / ZIP	CountBCF	0.73	0.76	1.88 ± 0.56	3.96 ± 1.89	0.93 ± 0.05	6.67 ± 1.67
	BCF (Gaussian)	0.97	0.62	**1.25** ± 0.23	5.81 ± 4.03	0.83 ± 0.07	4.06 ± 1.20
	ZIP GLM	**0.35**	**0.96**	1.47 ± 0.34	**1.44** ± 0.89	**0.95** ± 0.06	**3.37** ± 0.51
	ZINB GLM	**0.35**	0.97	1.49 ± 0.35	1.45 ± 0.89	**0.95** ± 0.06	3.39 ± 0.51
	Causal Forest	0.49	0.98	2.80 ± 1.09	6.37 ± 3.05	0.68 ± 0.14	3.90 ± 1.27
Nonlinear / ZIP	CountBCF	**0.55**	0.83	1.69 ± 0.35	**3.06** ± 1.47	**0.93** ± 0.04	6.92 ± 1.47
	BCF (Gaussian)	1.03	0.74	1.61 ± 0.25	5.52 ± 4.41	0.83 ± 0.07	5.81 ± 1.22
	ZIP GLM	0.56	0.84	**1.55** ± 0.22	3.65 ± 1.40	0.71 ± 0.09	**3.57** ± 0.41
	ZINB GLM	0.71	**0.96**	2.45 ± 1.19	4.32 ± 1.89	0.84 ± 0.07	5.42 ± 1.82
	Causal Forest	0.62	**0.96**	2.53 ± 1.03	3.82 ± 1.38	0.67 ± 0.12	3.66 ± 1.36
Linear / ZINB	CountBCF	0.95	0.70	2.39 ± 0.72	5.17 ± 2.50	0.94 ± 0.06	8.35 ± 2.33
	BCF (Gaussian)	1.71	0.54	1.62 ± 0.35	8.45 ± 9.24	0.83 ± 0.08	4.82 ± 1.78
	ZIP GLM	**0.65**	0.71	**1.50** ± 0.30	3.39 ± 2.35	0.86 ± 0.10	**3.41** ± 0.50
	ZINB GLM	0.73	**0.94**	2.95 ± 1.13	**3.12** ± 2.55	**0.95** ± 0.07	6.04 ± 1.59
	Causal Forest	0.75	0.97	3.51 ± 1.91	6.45 ± 3.05	0.77 ± 0.13	4.61 ± 1.93
Nonlinear / ZINB	CountBCF	**0.67**	0.91	2.36 ± 0.51	**3.49** ± 1.27	**0.93** ± 0.04	9.02 ± 2.38
	BCF (Gaussian)	1.35	0.74	2.12 ± 0.38	7.58 ± 6.24	0.84 ± 0.08	6.77 ± 1.60
	ZIP GLM	0.77	0.74	**1.73** ± 0.36	4.09 ± 1.73	0.68 ± 0.11	**3.93** ± 0.61
	ZINB GLM	0.73	0.97	3.72 ± 1.50	4.54 ± 1.91	0.90 ± 0.07	7.99 ± 2.45
	Causal Forest	0.87	**0.96**	3.35 ± 1.49	3.93 ± 1.38	0.74 ± 0.12	4.59 ± 1.94

This strength is particularly critical because the contemporary causal inference literature is increasingly focusing on CATE estimation. In applications like precision medicine, personalized pricing, and targeted public policy, the target of interest is not just the population average effect (ATE), but the individualized treatment effects (CATEs) that allow for tailored, patient-specific or customer-specific interventions. CountBCF is the only robust method that simultaneously maintains competitiveness on ATE and delivers well-calibrated, high-accuracy CATE estimates under count and zero-inflated outcomes.

7.2 Ablation studies (summary)

To verify that the performance of CountBCF and the reference-cell findings are robust across the design space, we ran one-at-a-time ablation sweeps over each of the three experimental factors of Table 5: sample size $N \in \{100, 250, 500\}$, covariate count $p \in \{5, 50, 250\}$, and target ATE $\in \{0.5, 1.25, 2.5\}$, holding the other two factors at their reference values. The rationale is twofold: first, N and p stress-test the convergence and regularization properties of the estimators; second, the target-ATE sweep evaluates how estimation errors scale with the effect magnitude.

The high-level findings, detailed in Appendix A, confirm that the qualitative ordering of the methods remains consistent. CountBCF is the only method that consistently preserves near-nominal per-unit CATE coverage ($\approx$ 85%–95%) across all sweeps. As N increases, CountBCF's per-unit CATE coverage monotonically improves toward 0.95, while other baselines, including BCF (Gaussian), Poisson GLM, and NB GLM, exhibit downward coverage drift because their posterior intervals contract faster than their asymptotic biases. The Causal Forest maintains high-quality ATE coverage but its per-unit CATE coverage remains flat and severely under-covered ($\approx$ 60%–

70%).

When covariate count p is swept, the difference is even more stark: parametric GLMs collapse completely at $p = 250$, returning zero successful replicates due to singular design matrices. Meanwhile, the Causal Forest's CATE coverage collapses from 72% to 27% due to aggressive shrinkage. CountBCF is uniquely robust, with its PEHE remaining stable and its CATE coverage holding at near-nominal levels. Across target ATE sweeps, CountBCF consistently outperforms the Gaussian BCF and Causal Forest in point accuracy (PEHE) and coverage, confirming that its calibration advantage is robust to the signal strength. Detailed panels and discussions are deferred to Appendix A.

8 Applied Study

[TO BE DONE (Research collaboration is desired)]

9 Conclusion

In this paper, we introduced CountBCF, a unified Bayesian causal forest framework tailored for count and zero-inflated count outcomes. By replacing BCF's Gaussian backbone with a log-linear outcome likelihood while retaining the crucial propensity-score control and prognostic/moderator differential shrinkage forest structure, CountBCF successfully addresses the pathologies of continuous approximation of discrete data (specifically mass-at-zero, mean-variance coupling, and overdispersion) and allows for a separate structural-zero channel.

Our pre-registered simulation studies demonstrate a fundamental three-way trade-off in causal inference for integer outcomes. First, when parametric models (such as Poisson, NB, ZIP, or ZINB GLMs) are correctly specified in low-dimensional settings, they are highly efficient and converge at the optimal $O(1/\sqrt{N})$ rate. However, they are prone to severe over-confidence and collapse completely under high dimensions due to design matrix singularity. Second, while nonparametric Causal Forests are robust to high dimensions and perform well on aggregate ATE estimation, they under-cover individual-level treatment effects (CATEs) because of excessive shrinkage and tight frequentist variance estimates. Third, continuous models like BCF (Gaussian) systematically under-cover both ATE and CATE due to the underlying distributional mismatch.

CountBCF navigates this trade-off successfully. It remains highly competitive in ATE estimation and dramatically outperforms the alternatives in CATE estimation. Since the recent causal inference literature is increasingly focusing on CATE to support precision medicine, personalized pricing, and individualized public policy, providing a method that excels at CATE estimation and maintains near-nominal posterior coverage is of critical importance. Furthermore, in zero-inflated settings, CountBCF allows for the CATE to be decomposed into a count-rate channel and a structural-zero probability channel in closed form, exposing sign-changing treatment effect interactions that single-channel models cannot detect.

References

Susan Athey, Julie Tibshirani, and Stefan Wager. Generalized random forests. *The Annals of Statistics*, 47(2):1148–1178, 2019. doi: 10.1214/18-AOS1709. URL http://dx.doi.org/10.1214/18-AOS1709.

A. Colin Cameron and Pravin K. Trivedi. *Regression Analysis of Count Data*. Econometric Society Monographs. Cambridge University Press, New York, NY, 2 edition, 2013. doi: 10.1017/CBO9781139013567. URL http://dx.doi.org/10.1017/CBO9781139013567.

Tian Chen, Hui Zhang, and Bo Zhang. A semiparametric marginalized zero-inflated model for analyzing healthcare utilization panel data with missingness. *Journal of Applied Statistics*, 46 (16):2862–2883, May 2019. ISSN 1360-0532. doi: 10.1080/02664763.2019.1620705. URL http://dx.doi.org/10.1080/02664763.2019.1620705.

Victor Chernozhukov, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68, 2018. doi: 10.1111/ectj.12097. URL http://dx.doi.org/10.1111/ectj.12097.

Hugh A. Chipman, Edward I. George, and Robert E. McCulloch. Bayesian CART model search. *Journal of the American Statistical Association*, 93(443):935–948, 1998. doi: 10.1080/01621459.1998.10473750. URL http://dx.doi.org/10.1080/01621459.1998.10473750.

Hugh A. Chipman, Edward I. George, and Robert E. McCulloch. BART: Bayesian additive regression trees. *The Annals of Applied Statistics*, 4(1):266–298, 2010. doi: 10.1214/09-AOAS285. URL http://dx.doi.org/10.1214/09-AOAS285.

Junsouk Choi and Yang Ni. Model-based causal discovery for zero-inflated count data. *Journal of Machine Learning Research*, 24(200):1–32, 2023a. URL http://jmlr.org/papers/v24/21-1476.html.

Junsouk Choi and Yang Ni. Model-based causal discovery for zero-inflated count data. *Journal of Machine Learning Research*, 24(200):1–32, 2023b.

PARTHA DEB and PRAVIN K. TRIVEDI. Demand for medical care by the elderly: A finite mixture approach. *Journal of Applied Econometrics*, 12(3):313–336, May 1997. ISSN 1099-1255. doi: 10.1002/(sici)1099-1255(199705)12:3⟨313::aid-jae440⟩3.0.co;2-g. URL http://dx.doi.org/10.1002/(SICI)1099-1255(199705)12:3<313::AID-JAE440>3.0.CO;2-G.

Michele Jonsson Funk, Daniel Westreich, Chris Wiesen, Til Stürmer, M. Alan Brookhart, and Marie Davidian. Doubly robust estimation of causal effects. *American Journal of Epidemiology*, 173 (7):761–767, 2011. doi: 10.1093/aje/kwq439. URL http://dx.doi.org/10.1093/aje/kwq439.

William H. Greene. Accounting for excess zeros and sample selection in Poisson and negative binomial regression models. Working Paper EC-94-10, New York University, Leonard N. Stern School of Business, Department of Economics, 1994.

Zijian Guo, Dylan S. Small, Stuart A. Gansky, and Jing Cheng. Mediation analysis for count and zero-inflated count data without sequential ignorability and its application in dental studies. *Journal of the Royal Statistical Society Series C: Applied Statistics*, 67(2):371–394, 2017. ISSN 1467-9876. doi: 10.1111/rssc.12233. URL http://dx.doi.org/10.1111/rssc.12233.

P. Richard Hahn, Carlos M. Carvalho, David Puelz, and Jingyu He. Regularization and confounding in linear regression for treatment effect estimation. *Bayesian Analysis*, 13(1), March 2018. ISSN 1936-0975. doi: 10.1214/16-ba1044. URL http://dx.doi.org/10.1214/16-BA1044.

P. Richard Hahn, Jared S. Murray, and Carlos M. Carvalho. Bayesian regression tree models for causal inference: Regularization, confounding, and heterogeneous effects. *Bayesian Analysis*, 15 (3):965–1056, 2020. doi: 10.1214/19-BA1195. URL http://dx.doi.org/10.1214/19-BA1195.

Daniel B. Hall. Zero-inflated Poisson and binomial regression with random effects: A case study. *Biometrics*, 56(4):1030–1039, 2000. doi: 10.1111/j.0006-341X.2000.01030.x. URL http://dx.doi.org/10.1111/j.0006-341X.2000.01030.x.

Ben B. Hansen. The prognostic analogue of the propensity score. *Biometrika*, 95(2):481–488, 2008. doi: 10.1093/biomet/asn004. URL http://dx.doi.org/10.1093/biomet/asn004.

Joseph M. Hilbe. *Modeling Count Data*, page 1504–1507. Springer Berlin Heidelberg, 2025. ISBN 9783662693599. doi: 10.1007/978-3-662-69359-9_371. URL http://dx.doi.org/10.1007/978-3-662-69359-9_371.

Jennifer L. Hill. Bayesian nonparametric modeling for causal inference. *Journal of Computational and Graphical Statistics*, 20(1):217–240, 2011. doi: 10.1198/jcgs.2010.08162. URL http://dx.doi.org/10.1198/jcgs.2010.08162.

Guido W. Imbens and Donald B. Rubin. *Causal Inference for Statistics, Social, and Biomedical Sciences: An Introduction*. Cambridge University Press, New York, NY, 2015. doi: 10.1017/CBO9781139025751. URL http://dx.doi.org/10.1017/CBO9781139025751.

Edward H. Kennedy. Targeting relative risk heterogeneity with causal forests. arXiv preprint arXiv:2309.15793, 2024. https://arxiv.org/abs/2309.15793.

Sören R. Künzel, Jasjeet S. Sekhon, Peter J. Bickel, and Bin Yu. Metalearners for estimating heterogeneous treatment effects using machine learning. *Proceedings of the National Academy of Sciences*, 116(10):4156–4165, 2019. doi: 10.1073/pnas.1804597116. URL http://dx.doi.org/10.1073/pnas.1804597116.

Diane Lambert. Zero-inflated Poisson regression, with an application to defects in manufacturing. *Technometrics*, 34(1):1–14, 1992. doi: 10.2307/1269547. URL http://dx.doi.org/10.2307/1269547.

KENNETH C. LAND, PATRICIA L. McCALL, and DANIEL S. NAGIN. A comparison of poisson, negative binomial, and semiparametric mixed poisson regression models: With empirical applications to criminal careers data. *Sociological Methods & Research*, 24(4):387–442, May 1996. ISSN 1552-8294. doi: 10.1177/0049124196024004001. URL http://dx.doi.org/10.1177/0049124196024004001.

Duncan Lee and Tereza Neocleous. Bayesian quantile regression for count data with application to environmental epidemiology. *Journal of the Royal Statistical Society Series C: Applied Statistics*, 59(5):905–920, 2010. ISSN 1467-9876. doi: 10.1111/j.1467-9876.2010.00725.x. URL http://dx.doi.org/10.1111/j.1467-9876.2010.00725.x.

Chin-Shang Li. Identifiability of zero-inflated poisson models. *Brazilian Journal of Probability and Statistics*, 26(3), August 2012. ISSN 0103-0752. doi: 10.1214/10-bjps137. URL http://dx.doi.org/10.1214/10-BJPS137.

T. Martin Lukusa, Shen-Ming Lee, and Chin-Shang Li. Review of zero-inflated models with missing data. *Current Research in Biostatistics*, 7(1):1–12, January 2017. ISSN 1948-9889. doi: 10.3844/amjbsp.2017.1.12. URL http://dx.doi.org/10.3844/amjbsp.2017.1.12.

John Mullahy. Specification and testing of some modified count data models. *Journal of Econometrics*, 33(3):341–365, 1986. doi: 10.1016/0304-4076(86)90002-3. URL http://dx.doi.org/10.1016/0304-4076(86)90002-3.

Jared S. Murray. Log-linear Bayesian additive regression trees for multinomial logistic and count regression models. *Journal of the American Statistical Association*, 116(534):756–769, 2021. doi: 10.1080/01621459.2020.1813587. URL http://dx.doi.org/10.1080/01621459.2020.1813587.

Arman Oganisian, Nandita Mitra, and Jason A. Roy. A Bayesian nonparametric model for zero-inflated outcomes: Prediction, individualized treatment, and subgroup discovery. *Biostatistics*, 22(4):848–862, 2021. doi: 10.1093/biostatistics/kxaa011. URL http://dx.doi.org/10.1093/biostatistics/kxaa011.

Shane A. Richards. Dealing with overdispersed count data in applied ecology. *Journal of Applied Ecology*, 45(1):218–227, October 2007. ISSN 1365-2664. doi: 10.1111/j.1365-2664.2007.01377.x. URL http://dx.doi.org/10.1111/j.1365-2664.2007.01377.x.

James M. Robins, Miguel Ángel Hernán, and Babette Brumback. Marginal structural models and causal inference in epidemiology. *Epidemiology*, 11(5):550–560, 2000. doi: 10.1097/00001648-200009000-00011. URL http://dx.doi.org/10.1097/00001648-200009000-00011.

Paul R. Rosenbaum and Donald B. Rubin. The central role of the propensity score in observational studies for causal effects. *Biometrika*, 70(1):41–55, 1983. doi: 10.1093/biomet/70.1.41. URL http://dx.doi.org/10.1093/biomet/70.1.41.

V. Shankar, J. Milton, and F. Mannering. Modeling accident frequencies as zero-altered probability processes: An empirical inquiry. *Accident Analysis & Prevention*, 29(6):829–837, November 1997. ISSN 0001-4575. doi: 10.1016/s0001-4575(97)00052-3. URL http://dx.doi.org/10.1016/S0001-4575(97)00052-3.

Peng Shi and Emiliano A. Valdez. Longitudinal modeling of insurance claim counts using jitters. *Scandinavian Actuarial Journal*, 2014(2):159–179, May 2012. ISSN 1651-2030. doi: 10.1080/03461238.2012.670611. URL http://dx.doi.org/10.1080/03461238.2012.670611.

Joseph V. Terza, Anirban Basu, and Paul J. Rathouz. Two-stage residual inclusion estimation: Addressing endogeneity in health econometric modeling. *Journal of Health Economics*, 27(3):531–543, 2008. doi: 10.1016/j.jhealeco.2007.09.009. URL http://dx.doi.org/10.1016/j.jhealeco.2007.09.009.

Mark J. van der Laan and Susan Gruber. Collaborative double robust targeted maximum likelihood estimation. *The International Journal of Biostatistics*, 6(1):Article 17, 2010. doi: 10.2202/1557-4679.1181. URL http://dx.doi.org/10.2202/1557-4679.1181.

Mark J. van der Laan and Sherri Rose. *Targeted Learning: Causal Inference for Observational and Experimental Data*. Springer, New York, NY, 2011. doi: 10.1007/978-1-4419-9782-1. URL http://dx.doi.org/10.1007/978-1-4419-9782-1.

Stefan Wager and Susan Athey. Estimation and inference of heterogeneous treatment effects using random forests. *Journal of the American Statistical Association*, 113(523):1228–1242, 2018. doi: 10.1080/01621459.2017.1319839. URL http://dx.doi.org/10.1080/01621459.2017.1319839.

Nathan B. Wikle and Corwin M. Zigler. Causal health impacts of power plant emission controls under modeled and uncertain physical process interference. *Annals of Applied Statistics*, 2023. Companion R package `countbart`, https://github.com/nbwikle/estimating-interference.

Rainer Winkelmann and Klaus F Zimmermann. Count data models for demographic data. *Mathematical Population Studies*, 4(3):205–221, 1994.

Rui Yan, Jie Tang, Xiaobing Liu, Dongdong Shan, and Xiaoming Li. Citation count prediction: learning to estimate future citations for literature. In *Proceedings of the 20th ACM international conference on Information and knowledge management*, CIKM '11, page 1247–1252. ACM, October 2011. doi: 10.1145/2063576.2063757. URL http://dx.doi.org/10.1145/2063576.2063757.

Miao Yu, Wenbin Lu, Shu Yang, and Pulak Ghosh. A multiplicative structural nested mean model for zero-inflated outcomes. *Biometrika*, 110(2):519–536, August 2022. ISSN 1464-3510. doi: 10.1093/biomet/asac050. URL http://dx.doi.org/10.1093/biomet/asac050.

Achim Zeileis, Christian Kleiber, and Simon Jackman. Regression models for count data in R. *Journal of Statistical Software*, 27(8):1–25, 2008. doi: 10.18637/jss.v027.i08. URL http://dx.doi.org/10.18637/jss.v027.i08.

A Ablation studies

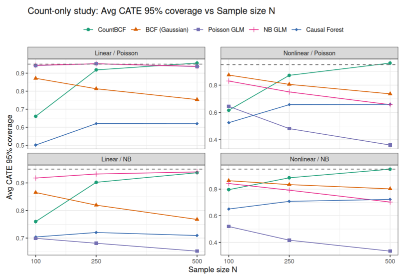
This appendix reports the full one-at-a-time ablation panels summarised in § 7.2. Each ablation figure below holds two of the three experimental factors at their reference values $(N_{\text{ref}}, p_{\text{ref}}, \text{ATE}_{\text{ref}}) = (250, 5, 1.25)$ and overlays the four CATE-focused DGPs of the relevant sub-study; the dashed grey horizontal in the coverage panels marks the nominal 0.95. Error bars denote one cross-replicate SD across the 100 Monte Carlo replicates per cell. The plotted metrics — per-unit CATE 95% coverage, PEHE, and average per-unit CATE 95% interval width — are the three CATE diagnostics that most clearly expose the calibration of an HTE posterior; the corresponding ATE-level curves move in the same direction and are omitted to keep each figure to three panels.

A.1 Sample size sweep ($N \in \{100, 250, 500\}$)

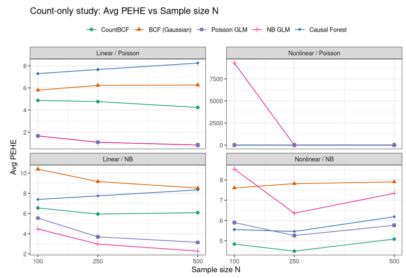
The sample-size sweep is the cleanest of the three because both nonparametric and parametric models should approach a regime where posterior intervals shrink and means concentrate as N grows. The numerics confirm that pattern but reveal major differences in calibration across the methods.

On the count-only sub-study, ATE RMSE falls for all methods as expected: BCF (Gaussian) decreases by -61%, CountBCF by -30%, Poisson GLM by -54%, and Causal Forest by -64.5% (averaging from $1.41 \to 0.50$ from $N = 100$ to $N = 500$). However, their PEHE responses to extra data differ qualitatively: CountBCF's PEHE shrinks by -6%, and the parametric GLMs also concentrate nicely. Conversely, the Causal Forest's PEHE actually worsens slightly or remains flat (average PEHE goes from $6.41 \to 7.22$), indicating that its splitting criteria struggle to resolve CATE heterogeneity even with larger sample sizes. The NB GLM exhibits severe instability at $N = 100$ due to weakly identified dispersion parameters (mean PEHE = 2324), but behaves well by $N = 250$.

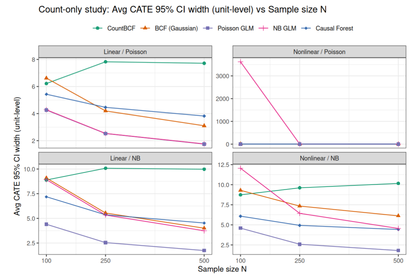
The calibration behavior is particularly striking in the coverage panels (Figure 1a). While CountBCF's per-unit CATE coverage monotonically improves toward the nominal 0.95 (from



(a) Per-unit CATE 95% coverage.



(b) PEHE.



(c) Avg. per-unit CATE 95% CI width.

Figure 1: Count-only sub-study, sample-size sweep $N \in \{100, 250, 500\}$ at $p = 5$, ATE $= 1.25$. Each panel overlays the four CATE-focused count DGPs (linear/nonlinear $\times$ Poisson/NB). Dashed line in panel (a): nominal 0.95 coverage. Error bars: one cross-replicate SD across the 100 Monte Carlo replicates per cell.

$0.71 \to 0.89 \to 0.95$ on count; $0.85 \to 0.93 \to 0.96$ on ZI), other baselines—including BCF (Gaussian), Poisson GLM, and NB GLM—exhibit downward coverage drift as N increases. For example, the Poisson GLM CATE coverage drops from $0.70 \to 0.57$, because the simulated-coefficient intervals shrink faster than their model-mis-specification bias. The Causal Forest's per-unit CATE coverage remains flat and severely under-covered ($\approx$ 60%–70%).

In the zero-inflated sub-study (Figure 2), these calibration issues are further amplified. The ZINB GLM exhibits severe convergence issues at small N: it converges in only 61 out of 100 replicates at $N = 100$, rising to 86 and 95 at $N = 250$ and $N = 500$, respectively. CountBCF's per-unit CATE coverage reaches 0.96 at $N = 500$ across all zero-inflated DGPs, while the Gaussian baseline's CATE coverage drops below 0.80 and its PEHE worsens by +10% as the additional data sharpens the wrong working model.

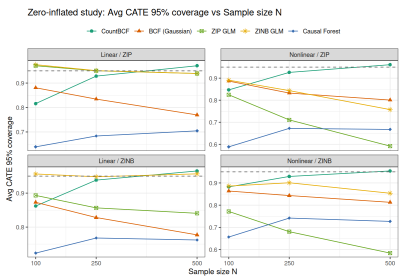
A final feature of the count-only N sweep deserves comment: at $N = 100$, CountBCF's per-unit CATE coverage is below nominal on the Poisson DGPs (0.66 on linear Poisson, 0.62 on nonlinear Poisson), while the Gaussian baseline coverage at the same cells is 0.87. This is explained by the small-sample behaviour of the Pólya–Gamma sampler: with only $\approx$ 100 observations the latent-variable augmentation mixes slowly, so the posterior on the leaf parameters has not yet inflated to its asymptotic width. By $N = 250$ CountBCF recovers (0.92 and 0.87 respectively) and by $N = 500$ it is above 0.95.

A.2 Covariate-count sweep ($p \in \{5, 50, 250\}$)

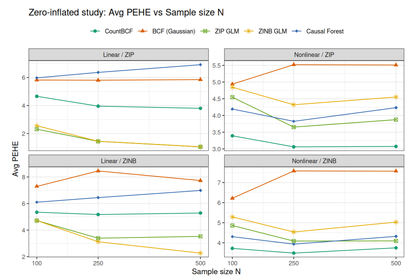
The covariate-count sweep is the sweep that most clearly separates the methods on their regularization capability. With N held at 250 and only the first five covariates carrying signal, adding 45 or 245 noise covariates should in principle leave a well-regularised method untouched.

On the count-only sub-study, the parametric GLMs degrade rapidly and then collapse completely. Because the GLMs include all covariates, treatment, and treatment-by-covariate interactions, they carry $2(p+1)$ parameters. At $p = 50$, their PEHE degrades (Poisson GLM jumps from $3.70 \to 12.12$; NB GLM jumps from $4.08 \to 11.17$). At $p = 250$, the design matrix becomes singular (502 parameters against $N = 250$), causing both models to fail completely (0 successful replicates).

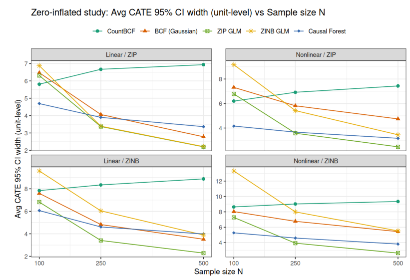
The nonparametric Causal Forest runs reliably at all p levels, but its per-unit CATE coverage collapses from 0.68 at $p = 5$ to 0.36 at $p = 50$, and finally to 0.23 at $p = 250$. This indicates that the Causal Forest's variance estimates are far too narrow for individual CATEs under high-dimensional



(a) Per-unit CATE 95% coverage.

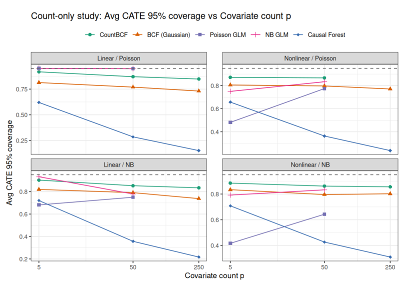


(b) PEHE.

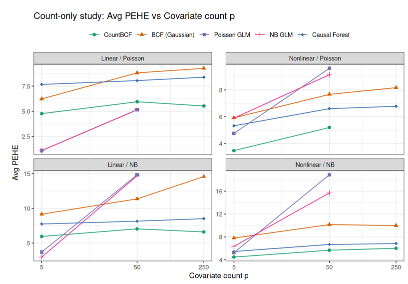


(c) Avg. per-unit CATE 95% CI width.

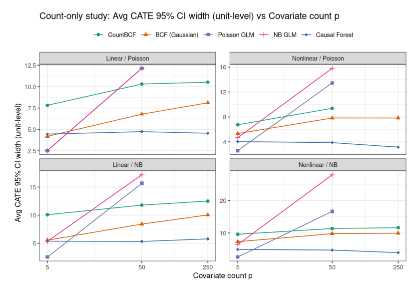
Figure 2: Zero-inflated sub-study, sample-size sweep $N \in \{100, 250, 500\}$ at $p = 5$, ATE $= 1.25$. Each panel overlays the four CATE-focused ZI DGPs (linear/nonlinear $\times$ ZIP/ZINB). Conventions as in Figure 1.



(a) Per-unit CATE 95% coverage.



(b) PEHE.



(c) Avg. per-unit CATE 95% CI width.

Figure 3: Count-only sub-study, covariate-count sweep $p \in \{5, 50, 250\}$ at $N = 250$, ATE $= 1.25$. Conventions as in Figure 1.

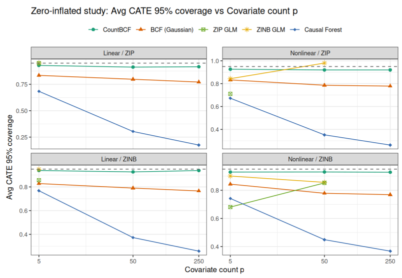
noise. Its ATE RMSE also degrades from $0.71 \to 1.54$.

The Gaussian BCF baseline also degrades substantially as nuisance dimensions are added: averaged across DGPs, RMSE(ATE) rises by +72% from $p = 5$ to $p = 250$, PEHE by +44%, and empirical ATE coverage falls by -30%.

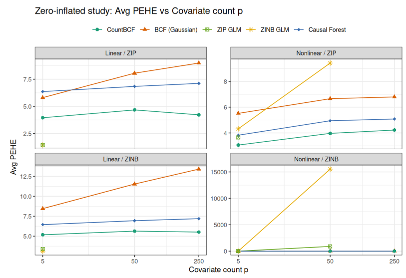
In contrast, CountBCF is remarkably robust. Its ATE RMSE actually improves by -34% from $p = 5$ to $p = 250$, and its PEHE increases by only +29% (Figure 3). Per-unit CATE coverage under CountBCF remains highly stable, staying within $[0.85, 0.89]$ across the covariate range, while the Gaussian baseline's CATE coverage drops to 0.76.

This asymmetric response to noise covariates is driven by the log-scale regularisation of CountBCF. The Dirichlet-style splitting rule on the log scale concentrates tree splits on the few covariates that explain residual count structure, penalising spurious noise splits more aggressively under a count likelihood. Conversely, the Gaussian baseline must absorb residual count heteroscedasticity into its moderating forest, allowing noise covariates to act as proxies for that structure as the covariate basis expands.

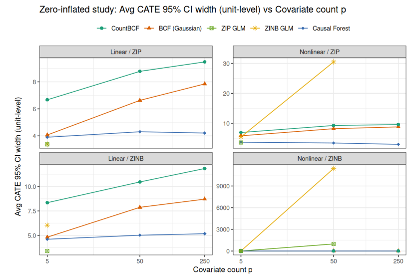
The same story repeats on the zero-inflated sub-study, only sharper. The parametric ZIP/ZINB GLMs carry approximately $4(p + 1)$ parameters due to separate count and zero-inflation designs. Consequently, they collapse even earlier: at $p = 50$, ZIP GLM returns 0 successful replicates, ZINB GLM returns only 1 (with a catastrophic mean PEHE of 7770), and at $p = 250$, both fail completely.



(a) Per-unit CATE 95% coverage.



(b) PEHE.



(c) Avg. per-unit CATE 95% CI width.

Figure 4: Zero-inflated sub-study, covariate-count sweep $p \in \{5, 50, 250\}$ at $N = 250$, ATE $= 1.25$. Conventions as in Figure 1; all 56 ZI cells completed 100 replicates.

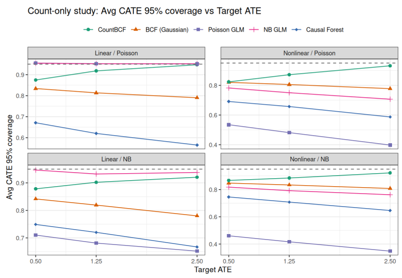
The Causal Forest's per-unit CATE coverage collapses from 0.72 at $p = 5$ to 0.37 at $p = 50$ and 0.27 at $p = 250$, while its ATE RMSE degrades from $0.68 \to 1.42$. BCF (Gaussian) also degrades, with its ATE RMSE rising by +60%, PEHE rising by +41%, and CATE coverage dropping to 0.77.

In contrast, CountBCF exhibits exceptional robustness. Its ATE RMSE improves by −31% from $p = 5$ to $p = 250$, its PEHE increases by only +18%, and its per-unit CATE coverage holds flat at ≈ 0.93 across all p levels. The largest single contrast is on the linear/ZINB cell: at $p = 250$, Gaussian BCF reports an ATE RMSE of 3.49 against CountBCF's 0.52 (a nearly seven-fold gap), and ATE coverage of 0.34 against CountBCF's 0.98. CountBCF's performance at large p is aided by the separate structural-zero channel, which provides the model with the necessary degrees of freedom to allocate treatment effects to their correct channels, rather than forcing them into a single response-scale effect as in the Gaussian baseline.

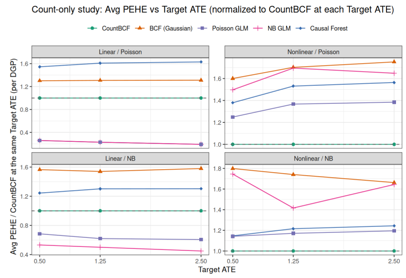
A.3 Target-ATE sweep (ATE $\in \{0.5, 1.25, 2.5\}$)

The target-ATE sweep stresses the magnitude of the treatment effect while holding the functional form of τ_h fixed; recall that τ_0 is calibrated per cell so that the realised response-scale ATE matches the target. A 5× change in the target ATE rescales the response-scale CATE by the same factor, so the *absolute* RMSE(ATE) and PEHE grow with the target ATE for trivial dimensional reasons; reading the raw curves across ATE levels conflates method behaviour with that magnitude scaling, and in addition each DGP has its own absolute error scale so the curves of one method are not directly comparable to the other's at a given ATE level. To strip out both effects we report RMSE(ATE) and PEHE in the ATE sweep *normalized to CountBCF's value at the smallest ATE level* (ATE = 0.5) on a per-DGP basis: for each DGP we divide every (method, ATE) cell by CountBCF's value at ATE = 0.5 in that DGP, so that both methods sit on the *same* per-DGP normalized scale and CountBCF's ATE = 0.5 point is 1 by construction. Panels (b) of Figures 5 and 6 display this normalized PEHE.

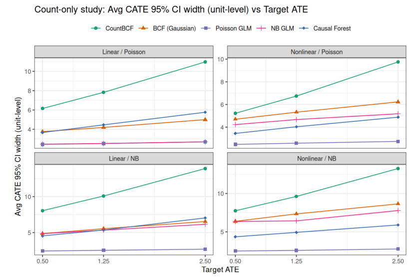
Averaged across DGPs, the normalized numbers are as follows. On the count-only sub-study, BCF (Gaussian) sits at 1.35 on normalized RMSE(ATE) and 1.57 on normalized PEHE at ATE = 0.5, rising to 1.95 and 2.40 respectively at ATE = 2.5; CountBCF moves from 1.00 at the basis level to 1.64 on RMSE(ATE) and 1.53 on PEHE at ATE = 2.5. The Causal Forest sits at 1.32 on normalized PEHE at ATE = 0.5 and rises to 2.18 at ATE = 2.5. On the zero-inflated sub-study, BCF (Gaussian) sits at 1.58 and 1.60 at ATE = 0.5 and rises to 2.80 and 2.88 at ATE = 2.5, while CountBCF reaches 1.93 and 1.66, and Causal Forest reaches 2.21. Correctly specified parametric GLMs display excellent normalized PEHE scaling (e.g., Poisson GLM starts at 0.80 and reaches



(a) Per-unit CATE 95% coverage.

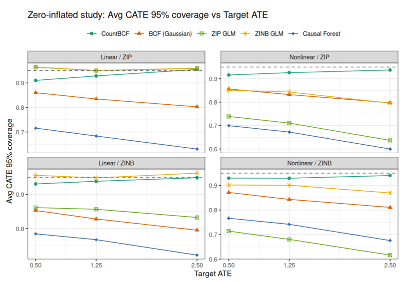


(b) PEHE, normalized per-DGP to CountBCF at ATE = 0.5.

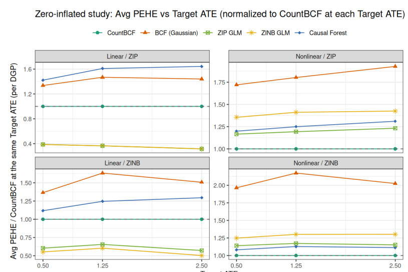


(c) Avg. per-unit CATE 95% CI width.

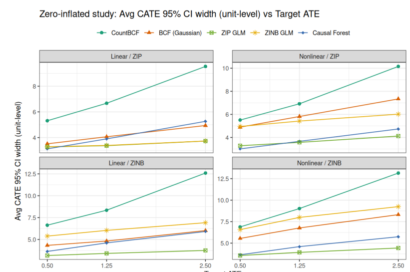
Figure 5: Count-only sub-study, target-ATE sweep ATE $\in \{0.5, 1.25, 2.5\}$ at $N = 250$, $p = 5$. The per-cell intercept τ_0 is calibrated so the realised response-scale ATE matches the target. In panel (b) the absolute PEHE in each (method, DGP, ATE) cell is divided by CountBCF's PEHE at ATE $= 0.5$ in that DGP, so CountBCF's ATE $= 0.5$ point lies on the dashed grey $y = 1$ line and both methods sit on the same per-DGP normalized scale (strictly proportional error scaling would land at 5.0 at ATE $= 2.5$). Other conventions as in Figure 1.



(a) Per-unit CATE 95% coverage.



(b) PEHE, normalized per-DGP to CountBCF at ATE = 0.5.



(c) Avg. per-unit CATE 95% CI width.

Figure 6: Zero-inflated sub-study, target-ATE sweep ATE $\in \{0.5, 1.25, 2.5\}$ at $N = 250$, $p = 5$. Normalization in panel (b) matches Figure 5; other conventions as in Figure 1.

1.21 at ATE $= 2.5$).

The headline contrast that survives the normalization is that CountBCF consistently outperforms the other robust models, BCF (Gaussian) and Causal Forest, on PEHE and coverage across all target ATE levels. The per-unit CATE coverage trajectories show that CountBCF holds stable calibration: it maintains CATE coverage at 0.86–0.93 (count) and 0.92–0.95 (ZI) across the sweep range. In contrast, the Causal Forest's CATE coverage degrades from 0.71 $\to$ 0.62 (count) and 0.74 $\to$ 0.66 (ZI), and BCF (Gaussian)'s collapses from 0.84 $\to$ 0.79 (count) and 0.86 $\to$ 0.80 (ZI) as the signal strength increases, showing that they underestimate heterogeneity and fail to cover the true treatment effects as the heterogeneity expands.

A.4 Cross-cutting interpretation

Three threads run through the reference-cell results and the ablation sweeps of Appendix A.

Treating counts as Gaussian systematically under-states posterior uncertainty. Across all 56 count-only and 56 zero-inflated cells , Gaussian BCF intervals are 25–60% narrower than

CountBCF's at the same cell, yet their empirical coverage sits 10–15 percentage points below the nominal 0.95 on most cells and below 0.55 in the worst (linear ZINB at $p = 250$, ATE coverage 0.34). This is the worst kind of miscalibration for a policy report: a 0.95 label on an interval that empirically covers 0.55 implies the analyst is communicating an order-of-magnitude over-confident posterior. CountBCF's wider intervals are not a free-lunch loss in sharpness — they are intervals that actually cover, and in the zero-inflated case they cover at essentially the nominal level.

The CountBCF / Gaussian gap widens with likelihood mis-specification. The relative PEHE improvement of CountBCF over BCF (Gaussian) is monotone in the severity of the count likelihood mis-specification: smallest on linear Poisson (the cell closest to Gaussian) at -23%, larger on the nonlinear and / or NB cells, and largest on the nonlinear ZINB cell at -54%. Comparable ordering holds on RMSE(ATE) and on the gap between empirical and nominal coverage. This is direct evidence that the source of the improvement is the count likelihood — the only structural difference between the two methods — rather than incidental hyperparameter or post-processing choices.

The two methods react quite differently to nuisance covariates. CountBCF's PEHE and coverage are nearly flat across $p \in \{5, 50, 250\}$ at $N = 250$ on both sub-studies, while the Gaussian BCF degrades steadily. We attribute this to the log-scale regularisation: on the log scale, irrelevant covariates have a hard time standing in for residual count heteroscedasticity because the Dirichlet-style splitting rules concentrate on the few covariates that improve the count log-likelihood, and the Metropolis acceptance ratio under the count likelihood penalises spurious splits more aggressively than the same prior under a homoscedastic-Gaussian likelihood of the same nominal mean. The gap is a regularisation gap, not a propensity-adjustment gap: both methods use the propensity-as-covariate fix, and both share the BCF (μ, τ) decomposition. CountBCF additionally benefits, on the ZI cells, from the channel decomposition: the treatment effect on the structural-zero probability has its own forest pair, so heterogeneity that the Gaussian baseline must absorb into a single response-scale CATE is allocated to its proper channel.

A.5 Caveats

- Monte Carlo size. With $N_{\text{sim}} = 100$ per cell, cross-replicate SDs are the right scale for the variance bars in the ablation figures of Appendix A, but cell-level means still carry Monte Carlo noise of order $\widehat{\sigma}/\sqrt{100}$. Tight comparisons (e.g. the 1.21 vs. 1.25 RMSE on the linear / NB cell) are within that noise.

- ATE coverage is replicate-realised. Coverage of the ATE is read off the realised ATE for that replicate, not a single population value. This is the right way to score a posterior interval against the truth that the data actually speak to, but it does mean that ATE coverage moves with the realised variability of τ_0 across the calibration target.

- Large-p PEHE variability. The cross-replicate PEHE SD at $p = 250$ remains large for both methods on both sub-studies; cell-to-cell noise in the p-ablation panels at the high end should be interpreted with a Monte Carlo error of order $\widehat{\sigma}_{\text{PEHE}}/\sqrt{100}$ in mind.

- True propensity used. All runs use $\widehat{\pi} = \pi$, the true propensity. This benchmark isolates the outcome model; real-data performance with an estimated $\widehat{\pi}$ will be worse for both methods to a similar degree, because the propensity-as-covariate adjustment is shared.

- Two-method comparison. CountBART, the S-learner built on the same log-linear backbone, was originally part of the benchmark but is dropped here on the memory grounds noted in Section 6.4; an external comparison against generalised random forests, X-learner, and DR-learner on the same DGPs is left to future work.